Cyclic dominance of a Potts–RPS model on complex networks

Full test and results report

Reproducible run 2026-07-02 · 18/18 steps OK · all figures + data regenerated byte-for-byte identical (deterministic seeds)

0. Model, notation and definitions

Agents sit on the nodes of a network and each holds one strategy in {Rock, Paper, Scissors} — a q=3 Potts spin. Interactions reward matching your neighbours (drives *order*) but also contain the cyclic RPS dominance (drives endless *cycling*); a single knob ε sets their ratio. The question throughout: for which (ε, network) does the population reach consensus, and when does it chase itself forever?

The model	
agent / node	one site of the network; holds strategy $s_i \in$ {R, P, S} and interacts only with its graph neighbours
payoff matrix	$P = I + \varepsilon\,\text{skew}$: the identity part pays for matching neighbours (ferromagnetic, orders); the antisymmetric part encodes the cycle *Paper beats Rock, Scissors beats Paper, Rock beats Scissors*
ε	cyclic-dominance strength ($0 \le \varepsilon \le 1$), the control parameter of every sweep
utility U_i	sum of P-payoffs of i's strategy against each neighbour's; scales with degree k, so connectivity acts like an inverse temperature
Glauber update	a node adopts a proposed strategy with probability $1/(1+e^{-\Delta U/T})$; temperature T=0.65 everywhere in this report
sweep, burn-in	one sweep = N attempted single-node updates (the MC time unit); burn-in = initial sweeps discarded before measuring
seed	RNG / graph-realisation seed; curves are averaged over seeds where stated
Networks	
ER	Erdős–Rényi random graph: edges placed independently; narrow, homogeneous degree distribution around $\langle k\rangle$
BA	Barabási–Albert graph: grown by preferential attachment; heavy-tailed $P(k)$ with *hubs* (nodes of very high degree)
N, $\langle k\rangle$, $P(k)$	number of nodes; average degree; degree distribution
Observables	
(r, p, s)	global fractions of the population playing Rock, Paper, Scissors
ψ	$r + p\,e^{i2\pi/3} + s\,e^{i4\pi/3}$: complex sum of the three fractions; large $\lvert\psi\rvert \Leftrightarrow$ one strategy dominates
m_ψ	$\lvert\langle\psi\rangle_t\rvert$, the time-averaged order parameter: $\to 1$ consensus (*ordered phase*), $\to 0$ when strategies chase each other (*cycling phase*); computed by pipeline steps P2–P4 of Sec. 0.1
ε_c	critical cyclic strength where order gives way to cycling; quoted at the interpolated m_ψ=0.5 crossing throughout (pipeline step P6)
conversion	fraction of *free* (non-zealot) nodes playing the zealots' strategy
Methods	
MC	Monte Carlo: stochastic agent-level simulation on the actual graph — the ground truth
HMF	homogeneous mean field: three coupled ODEs for (r, p, s) in which every node feels the same mean degree $\langle k\rangle$
DMF	degree-based mean field: one (r, p, s) per degree class, coupled through $P(k)$ — resolves degree heterogeneity
RMSE	root-mean-square error of a mean-field $m_\psi(\varepsilon)$ curve against the MC curve over the sweep grid
FSS	finite-size scaling: how the transition sharpens and its apparent ε_c shifts as N grows — the signature of a true phase transition
Perturbations	
zealot	a node locked to one strategy forever: it influences neighbours but never updates; z = zealot fraction
defect	quenched (frozen-in) damage: a fraction f of edges removed (broken links) or nodes removed (vacancies)
effective $\langle k\rangle$	the mean degree of the network *after* damage

fixed point / limit cycle	steady state vs closed periodic orbit of the mean-field dynamics; the ternary *simplex* is the triangle of all (r, p, s) compositions, corners = pure strategies

0.1 How every number is obtained — the measurement pipeline

Every m_ψ, strategy fraction and ε_c in this report is produced by one fixed chain from raw strategies to the printed digit. The steps are numbered (P1–P6) so each result section below can state exactly which it uses.

(P1) Dynamics generate the microstate. Nodes start from uniformly random strategies. Each sweep makes N single-node update attempts: pick a random node i, propose one of the two other strategies s', compute the utility change against i's neighbours only, $\Delta U = \sum_{j\in\partial i}\left[P_{s'\,s_j} - P_{s_i\,s_j}\right]$ with $P = I + \varepsilon\,\text{skew}$, and accept with the Glauber probability $1/(1 + e^{-\Delta U/T})$. Zealots are skipped (they never update) but still appear in their neighbours' ΔU. This runs for the stated number of sweeps; the first *burn-in* sweeps are discarded unmeasured (default 30%).

(P2) Count the three strategy clusters. At the end of every post-burn-in sweep the engine counts the population of each strategy cluster — $N_R(t)$, $N_P(t)$, $N_S(t)$, the number of nodes currently playing Rock, Paper, Scissors — and normalises to fractions $r(t) = N_R(t)/N$, $p(t) = N_P(t)/N$, $s(t) = N_S(t)/N$, with $r + p + s = 1$. These three time series are the *only* raw measurement; everything below is computed from them.

(P3) Map the fractions to one complex number. The three strategies are assigned unit vectors 120° apart in the complex plane, and the fractions are summed along them:

$$\psi(t) = r(t) + p(t)\,e^{i2\pi/3} + s(t)\,e^{i4\pi/3} = \underbrace{r - \tfrac{1}{2}(p+s)}_{\operatorname{Re}\psi} + i\,\underbrace{\tfrac{\sqrt{3}}{2}(p-s)}_{\operatorname{Im}\psi}.$$

Geometry does the classification: if one cluster dominates, ψ sits near that strategy's corner and $|\psi| \to 1$; the symmetric mix $r = p = s = \frac{1}{3}$ gives $\psi = 0$; and RPS cycling makes $\psi(t)$ *rotate* around the origin at roughly constant radius.

(P4) Time-average, then take the modulus. The order parameter is

$$m_\psi = \Big|\,\frac{1}{M}\sum_{t>t_{\text{burn}}}\psi(t)\,\Big|$$

over the M measurement sweeps. The order of operations is the whole trick: averaging the *vector* first means a rotating ψ cancels itself and cycling scores $m_\psi \approx 0$, while static consensus keeps $m_\psi \approx 1$. (Taking $|\psi(t)|$ first would score both phases high.) The identical estimator is implemented in `common/observables.py` (Python MC, HMF, DMF) and `drivers/mc_engine.cpp` (C++).

(P5) Average over seeds. Where a section says "n seeds", steps P1–P4 are repeated on n independent graph realisations (graph seed = RNG seed = $1\ldots n$) and the resulting curves are averaged point-by-point; tables show the seed-averaged values.

(P6) Extract ε_c by interpolation. A transition point is read off a sweep $m_\psi(\varepsilon)$ by sorting in ε, finding the first grid point with $m_\psi < 0.5$, and linearly interpolating between the two bracketing points: $\varepsilon_c = \varepsilon_0 + (0.5 - m_0)\,(\varepsilon_1 - \varepsilon_0)/(m_1 - m_0)$. If the curve never drops below 0.5 the last grid point is reported. One convention everywhere (`phase_diagram/critical_boundary.py`; this report's tables use the same function).

Derived quantities (each computed inside the same P2–P5 measurement window): *conversion* = per-sweep (free nodes playing the zealot strategy) / (all free nodes), time- then seed-averaged — zealots are excluded from numerator *and* denominator, so it measures genuine influence; ρ_x = time-averaged global fraction of cluster x (zealots included, i.e. P2 averaged); *RMSE* $= \sqrt{\frac{1}{n}\sum_j\left(m_j^{\text{MF}} - m_j^{\text{MC}}\right)^2}$ over the n points of the ε grid; *effective* $\langle k\rangle = 2E'/N'$ of a damaged graph (surviving edges E', surviving nodes N'), averaged over damage realisations; *max|slope|* $= \max_j |m_{j+1} - m_j|/(\varepsilon_{j+1} - \varepsilon_j)$, the discrete steepness of a transition curve.

Robustness protocol. Each results section below ends with one or more *Robustness* subsections that audit the parameters that section holds fixed. Every such experiment follows the same discipline: identify the fixed parameters, state a falsifiable hypothesis (mechanism, which metrics should move, which should not) *before* running, sweep the parameter over a grid, then confront the data with the hypothesis — including the cases where a parameter is expected *not* to matter, which are tested, not assumed. The pre-registered hypotheses are in `sensitivity/HYPOTHESES.md`, the full analysis in `sensitivity/FINDINGS.md`; every number in these subsections is computed at build time from the `sensitivity/*.csv` tables, through the same pipeline P1–P6.

1. Run manifest and reproducibility

Every step below was produced in one clean run of `./run_all.sh` (clean-rebuild C++ engine → validation test → all simulations). Per-step logs live in `logs/`; the machine-readable manifest is `logs/manifest.csv`.

step	status	wall (s)
test_validate_engine	OK	4
mean_field_hmf	OK	1
mean_field_sweep	OK	7
mean_field_compare_suite_ER	OK	5
mean_field_compare_suite_BA	OK	6
monte_carlo_mc	OK	2
monte_carlo_compare	OK	77
phase_diagram_ER	OK	6
phase_diagram_BA	OK	6
phase_diagram_boundary	OK	0
dynamics_ternary	OK	2
dynamics_fss	OK	3
dynamics_stability	OK	1
zealots_experiment	OK	2
zealots_experiment_hubs	OK	4
zealots_experiment_mixed	OK	2
defects_experiment	OK	7
defects_collapse	OK	1
total	**18/18 OK**	**136**

Table 2: All simulation steps, status and wall-clock time (16-core host, C++20 engine `-O3 -march=native`). After the run `git status` reports zero changes to any tracked figure/CSV: fully deterministic.

2. Engine validation (C++ vs pure-Python MC)

What it is. A correctness cross-check: the same graph is evolved by the fast C++ engine and by an independent pure-Python Monte-Carlo reference, at four values of ε spanning both phases. Because the two use different RNG streams, agreement is expected in *regime* (ordered vs cycling), not in the last digit. This licenses using the C++ engine for everything below. Same graph through both engines; independent RNG streams, so agreement is in phase not last digit (log: `test_validate_engine.log`).

Parameters — each defined.

- N=500 — number of nodes (system size).
- $\langle k\rangle$=10 — target average degree: the mean number of neighbours per node the ER construction aims for.
- graph seed = 42 — RNG seed of the graph construction; fixes the one ER realisation that both engines share.
- T=0.65 — Glauber temperature: the noise scale in the acceptance probability $1/(1+e^{-\Delta U/T})$; higher T = more random flips.
- sweeps = 1500 — simulation length; one sweep = N attempted single-node updates (the MC time unit).
- burn-in = 30% — initial transient discarded before measuring (Python: first 30% of the recorded series; C++: the first 450 sweeps).
- MC seed = 7 — RNG seed of the update dynamics; the two engines use different RNG algorithms, so their random streams are independent even at equal seed.
- $\varepsilon \in \{0.2, 0.5, 0.7, 0.9\}$ — cyclic-dominance strengths probed: two expected in the ordered phase, two in the cycling phase.

How the numbers are obtained. Each m_ψ cell is a single run through pipeline P1–P4 (no seed averaging). The verdict is *agree* iff both engines land on the same side of $m_\psi = 0.5$, i.e. classify the same phase; the speedup is the ratio of the two total wall-clock times parsed from the validation log.

ε	Python m_ψ	C++ m_ψ	regime / verdict
0.2	0.999	0.999	agree (ordered)
0.5	0.991	0.992	agree (ordered)
0.7	0.004	0.007	agree (cycling)
0.9	0.004	0.003	agree (cycling)

Table 3: Engine agrees in every regime; measured speedup **31**× (Python 3.27 s vs C++ 0.10 s for the 4-point sweep; timing varies run-to-run, the m_ψ values do not).

2.1 Robustness: does the agreement hold across the parameter space?

What it is. The table above is 4 ε values on one graph with one seed pair. Here the same paired comparison (identical graph and seed number through both engines) is repeated over a grid of 45 pairs: $N \times \varepsilon \times$seed.

Hypothesis — stated before running. The two engines implement the same stochastic process with independent RNG streams, so they must agree in *distribution*, not per run: same side-of-0.5 verdicts everywhere except possibly where $m_\psi \approx 0.5$ (inside the narrow transition window); $|m_{py} - m_{cpp}|$ should peak near the transition and shrink as $1/\sqrt{N}$ (self-averaging of a global average). A systematic offset at any point away from the transition would indicate an implementation difference, not noise.

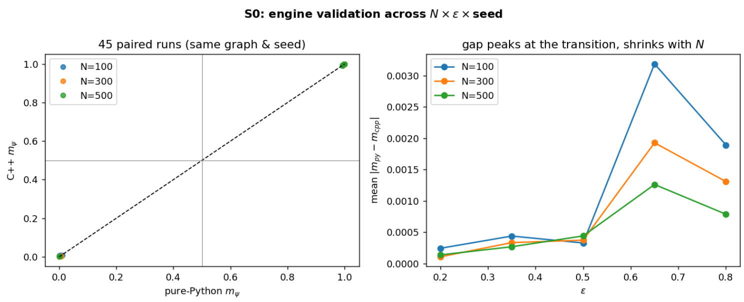
Parameters — each defined.

- $N \in \{100, 300, 500\}$ — system sizes; tests the predicted $1/\sqrt{N}$ shrinkage of the engine-to-engine gap.
- $\varepsilon \in \{0.2, 0.35, 0.5, 0.65, 0.8\}$ — spans deep order, the transition neighbourhood and deep cycling at $\langle k \rangle$=10.
- seeds $\in \{7, 8, 9\}$ — both the pure-Python and the C++ run use the same seed number but independent RNG algorithms, so per-pair equality is not expected — only statistical agreement.
- $\langle k \rangle$=10, T=0.65, 1500 sweeps, burn-in 30%, graph seed 42 — identical to Sec. 2, so the grid extends the headline validation rather than replacing it.

How the numbers are obtained. Each pair is one P1–P4 run per engine on the same edge list; the verdict criterion and $|m_{py} - m_{cpp}|$ are computed per pair, then summarised per N. Data: `sensitivity/sens_validation.csv`.

N	max $\lvert\Delta m_\psi\rvert$	mean $\lvert\Delta m_\psi\rvert$
100	0.0054	0.0012
300	0.0049	0.0008
500	0.0018	0.0006

Verdicts: **45/45** pairs classify the same phase. The predicted transition-window disagreements never materialised because no test point landed inside the (narrower than expected) window.



Verdict vs hypothesis. Confirmed, and stronger than hypothesised: 45/45 verdicts agree, the maximum gap is 0.0054, and the per-N maxima fall monotonically with N (0.0054 / 0.0049 / 0.0018) — the $1/\sqrt{N}$ self-averaging signature. No systematic offset anywhere: the C++ engine is the same physics as the reference, across the space actually used in this report, not just at the 4 showcase points.

3. Mean field and Monte Carlo

3.1 HMF ε-sweep: connectivity stabilises order

What it is. The homogeneous mean field replaces the network by three coupled ODEs for the population fractions (r, p, s), in which every node feels the same mean degree $\langle k \rangle$. We integrate to steady state and sweep the cyclic-dominance strength ε at several $\langle k \rangle$, recording m_ψ. It is the cheapest probe of the order→cycling transition and how the critical point ε_c depends on connectivity.

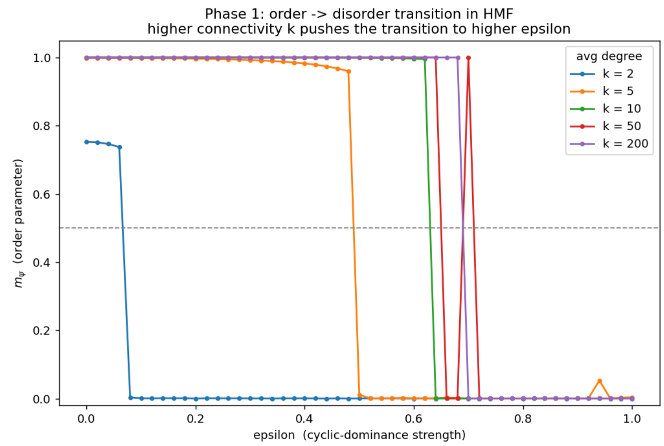
Parameters — each defined.

- steps = 4000 — iterations of the deterministic HMF recursion (its time axis, replacing MC sweeps).
- $(r, p, s)_0 = (0.40, 0.35, 0.25)$ — initial composition of the population; deliberately asymmetric so the dynamics is not stuck on the unstable symmetric point $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.
- T=0.65 — Glauber temperature entering the mean-field transition rates.
- measurement window = last 50% — the part of the trajectory that P4 averages into m_ψ (the first half is the burn-in of the map).
- ε: 51 points on $[0, 1]$ — resolution of the control-parameter grid (spacing 0.02).
- $\langle k \rangle \in \{2, 5, 10, 50, 200\}$ — mean degree multiplying the utilities; the *only* place network structure enters the HMF.
- ε_c — interpolated m_ψ=0.5 crossing of each column (P6).

How the numbers are obtained. P1 is replaced by the deterministic HMF map (every node feels the mean mix, so the (r, p, s) update is a closed 3-variable recursion); each iterate *is* the global fraction vector, so P2 is trivial. m_ψ then follows P3–P4 over the last 50% of the 4000-step trajectory, and each ε_c in the table applies P6 to that column's 51-point $m_\psi(\varepsilon)$ curve.

$\langle k \rangle$	2	5	10	50	200
ε_c	0.07	0.49	0.63	0.65	0.69

ε_c rises monotonically with mean degree $\langle k \rangle$: denser networks sustain order against stronger cyclic pressure.



3.2 MC vs HMF vs DMF: RMSE against ground truth ($\langle k \rangle$=10)

What it is. A three-way accuracy test. We compute $m_\psi(\varepsilon)$ three ways on the same $\langle k \rangle$=10 setting: full agent-level **MC** (the ground truth), **HMF** (one mean degree), and **DMF** (degree-based mean field, one ODE class per degree, using the full $P(k)$). We score each mean field by its root-mean-square error against MC across the sweep. The question: does resolving the degree distribution (DMF) buy accuracy, and does that gain grow on a heterogeneous BA graph with hubs?

Parameters — each defined.

- graphs — one ER and one BA realisation, N=800 nodes, target $\langle k \rangle$=10, graph seed 1: a single fixed network per type so all three methods describe the *same* object.
- MC (ground truth) — C++ engine: T=0.65 (Glauber temperature), 1500 sweeps (one sweep = N update attempts), burn-in 450 (discarded), engine seed 1.
- HMF input: k = measured $\langle k \rangle$ of that graph — the mean field sees the network only through this one number.
- DMF input: measured degree histogram $P(k)$ of that graph — one (r, p, s) per degree class, so degree heterogeneity survives.
- mean-field steps = 4000 — iterations of both deterministic maps; m_ψ from the last 50%.
- ε: 26 points on $[0, 1]$ (spacing 0.04) — the common grid on which all three curves are computed; RMSE is taken over all 26 points.

How the numbers are obtained. Three $m_\psi(\varepsilon)$ curves per graph: MC by P1–P4 (one engine run per ε), HMF/DMF by the deterministic maps fed the *measured* $\langle k \rangle$ / $P(k)$ of the same graph, then P3–P4. Each RMSE cell is $\sqrt{\frac{1}{26}\sum_j (m_j^{\mathrm{MF}} - m_j^{\mathrm{MC}})^2}$; DMF gain = RMSE(HMF) − RMSE(DMF); ε_c columns by P6.

graph	RMSE(HMF)	RMSE(DMF)	DMF gain	ε_c(MC)	ε_c(MF)
ER	0.338	0.334	**0.004**	0.50	0.62
BA	0.336	0.328	**0.008**	0.50	0.62

Table 4: DMF beats HMF on both graphs; the DMF advantage is ∼2.2× larger on heterogeneous BA. Both mean fields overestimate the ordered phase (MC turns over at ε≈0.50 vs ≈0.62 on ER).

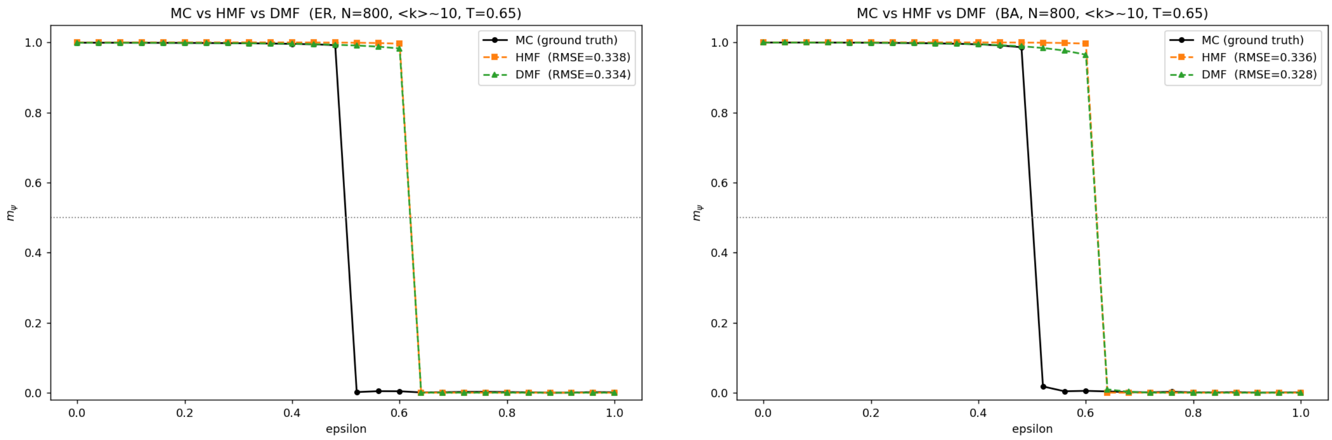


Figure 1: Comparison suite, ER (left) and BA (right).

The same test across T, $\langle k\rangle$ and N. The single operating point above could flatter (or hide) a mean field. The identical protocol is repeated over a grid: one axis varied at a time around the base point (T=0.65, $\langle k\rangle$=10, N=800), on both graphs.

Hypothesis — stated before running. RMSE(DMF) $\leq$ RMSE(HMF) everywhere, with the larger DMF advantage on BA (only the DMF sees the heavy-tailed $P(k)$). RMSE should *grow* with T (the MC boundary slides while the mean field lags), *shrink* with $\langle k\rangle$ (MC ε_c rises toward the mean-field value), and be nearly flat in N (both mean fields are N-blind, and the MC curve is intensive).

Parameters — each defined.

- $T \in \{0.4, 0.65, 1.0\}$, $\langle k\rangle \in \{6, 10, 20\}$, $N \in \{400, 800, 1600\}$ — one axis at a time around the base point (7 distinct cells).
- graphs — one ER and one BA per cell; MC ground truth = C++ engine, 2 graph seeds averaged (P5), 26 ε points on $[0, 1]$.
- HMF/DMF inputs — measured $\langle k\rangle$ / measured $P(k)$ of each graph, T passed through, 4000 map steps: identical to the base experiment.

How the numbers are obtained. Per cell and graph: three $m_\psi(\varepsilon)$ curves (P1–P4 for MC, the maps for HMF/DMF), RMSE per the Sec. 0.1 definition, ε_c by P6. Data: `mean_field/compare_grid.csv`.

axis	value	ER: RMSE(HMF)/RMSE(DMF)		BA: RMSE(HMF)/RMSE(DMF)	
T	0.4	0.392	0.391	0.390	0.388
T	0.65 (base)	0.338	0.334	0.298	0.289
T	1	0.272	0.324	0.273	0.255
$\langle k\rangle$	6	0.384	0.369	0.381	0.352
$\langle k\rangle$	10 (base)	0.338	0.334	0.298	0.289
$\langle k\rangle$	20	0.276	0.276	0.276	0.276
N	400	0.275	0.289	0.289	0.281
N	800 (base)	0.338	0.334	0.298	0.289
N	1600	0.338	0.334	0.337	0.329

Table 5: Mean-field accuracy across the grid: DMF $\leq$ HMF in 12/14 cells.

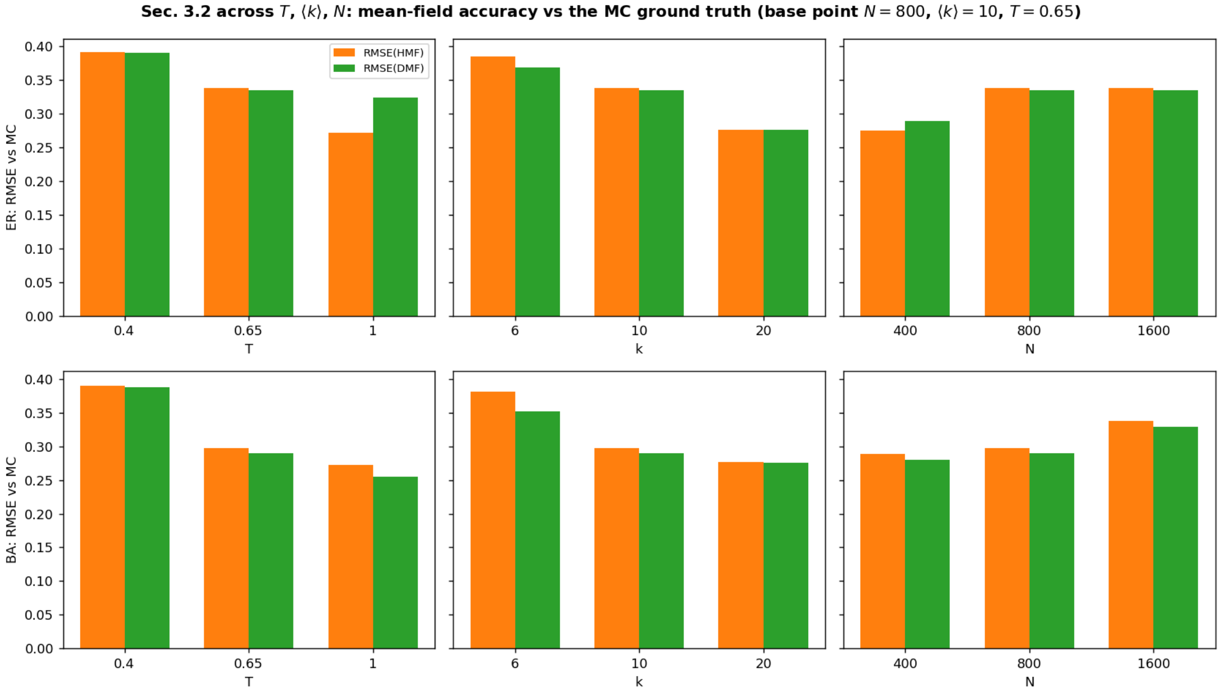


Figure 2: RMSE vs MC per axis (top: ER, bottom: BA).

Verdict vs hypothesis. DMF $\leq$ HMF holds in 12/14 cells — 7/7 on BA vs 5/7 on ER: the exceptions are ER cells, where $P(k)$ is narrow so the DMF's extra structure buys nothing and seed noise decides — exactly the pattern the "DMF wins *via* $P(k)$" mechanism implies. The $\langle k \rangle$ trend is as predicted (RMSE falls 0.38 → 0.28 on BA as the MC boundary climbs toward the mean-field one). The T hypothesis is *wrong in sign*: RMSE falls with T (0.39 → 0.27 on BA) because at $\langle k \rangle$=10 the HMF is still T-responsive — its ε_c drops (0.70 → 0.54) *faster* than the MC one (0.54 → 0.46), narrowing the mismatch region; the "T-blind mean field" picture only sets in at high $\langle k \rangle$ (Sec. 4.2). And N is *not* the null axis it was assumed to be: RMSE grows with N (0.29 → 0.34 on BA) because the MC ε_c slides down as $1/N$ (0.52 → 0.50, the first-order drift of Sec. 5.1) while the mean fields stand still — an independent MC-side confirmation of that drift.

3.3 Direct MC vs HMF overlay (ER, $\langle k \rangle$=10)

What it is. A head-to-head on a single ER graph: the MC transition curve overlaid on the HMF prediction, to see exactly where and by how much the mean-field approximation misplaces the critical point on a finite network.

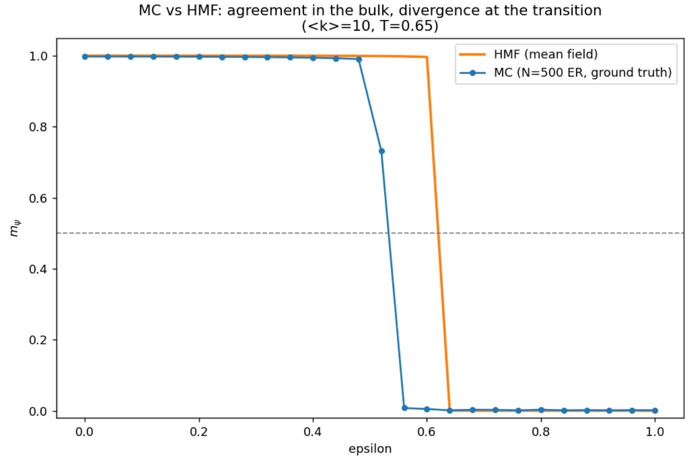
Parameters — each defined.

- graph — ER, N=500 nodes, $\langle k \rangle$=10 (average degree), graph seed 1 (fixes the realisation).
- MC — pure-Python reference implementation: T=0.65 (Glauber temperature), 1200 sweeps, burn-in 30% (first 360 sweeps discarded).
- MC seeds $\{1, 2, 3\}$ — three independent RNG streams for the dynamics; the plotted curve is their average (P5).
- HMF — k=10 fed to the map, 4000 steps, m_ψ from the last 50%.
- ε: 26 points on $[0, 1]$ — shared sweep grid for both curves.

How the numbers are obtained. MC row: P1–P4 per ε on each of the three seeds, then the seed average (P5); HMF row from the map as in Sec. 3.1. Both ε_c values by P6 on the respective 26-point curve.

model	ε_c
MC (ground truth)	0.53
HMF	0.62

MC transition precedes HMF: the mean field overestimates the ordered region, as expected on a finite network.



The same overlay across T, $\langle k\rangle$, N — and on BA. One panel is one anecdote. The overlay is repeated with the validated C++ engine (Secs. 2–2.1) as the MC — which is what makes a 12-panel grid affordable — varying one axis at a time around (ER, T=0.65, $\langle k\rangle$=10, N=500), plus a BA row.

Hypothesis — stated before running. The mean field misplaces the transition *systematically*, not randomly: the MC–HMF gap in ε_c should shrink with $\langle k\rangle$ (approaching the high-k regime of Sec. 4.2), persist at every T (both boundaries move down together at this degree, cf. Sec. 3.2), and in N grow slowly as the MC crossing slides down ($1/N$, Sec. 5.1) under a fixed HMF curve. The BA overlays should reproduce the ER ones at matched $\langle k\rangle$ (Sec. 4.1: ER $\approx$ BA).

Parameters — each defined.

- rows — vary $T \in \{0.4, 0.65, 1.0\}$; vary $\langle k\rangle \in \{6, 10, 20\}$; vary $N \in \{250, 500, 1000\}$; BA row: BA at $\langle k\rangle$=10 and 20, plus a direct ER-vs-BA MC panel.
- MC — C++ engine, 3 graph seeds averaged (P5), 26 ε points; engine seed = graph seed; production sweeps/burn-in.
- HMF — fed the *measured* $\langle k\rangle$ of each graph (seed-averaged), 4000 map steps; T passed through.

How the numbers are obtained. Per panel: P1–P5 for the MC curve, the map for HMF, both ε_c by P6 (printed in each panel title). Data: `monte_carlo/mc_vs_hmf_grid.csv`.

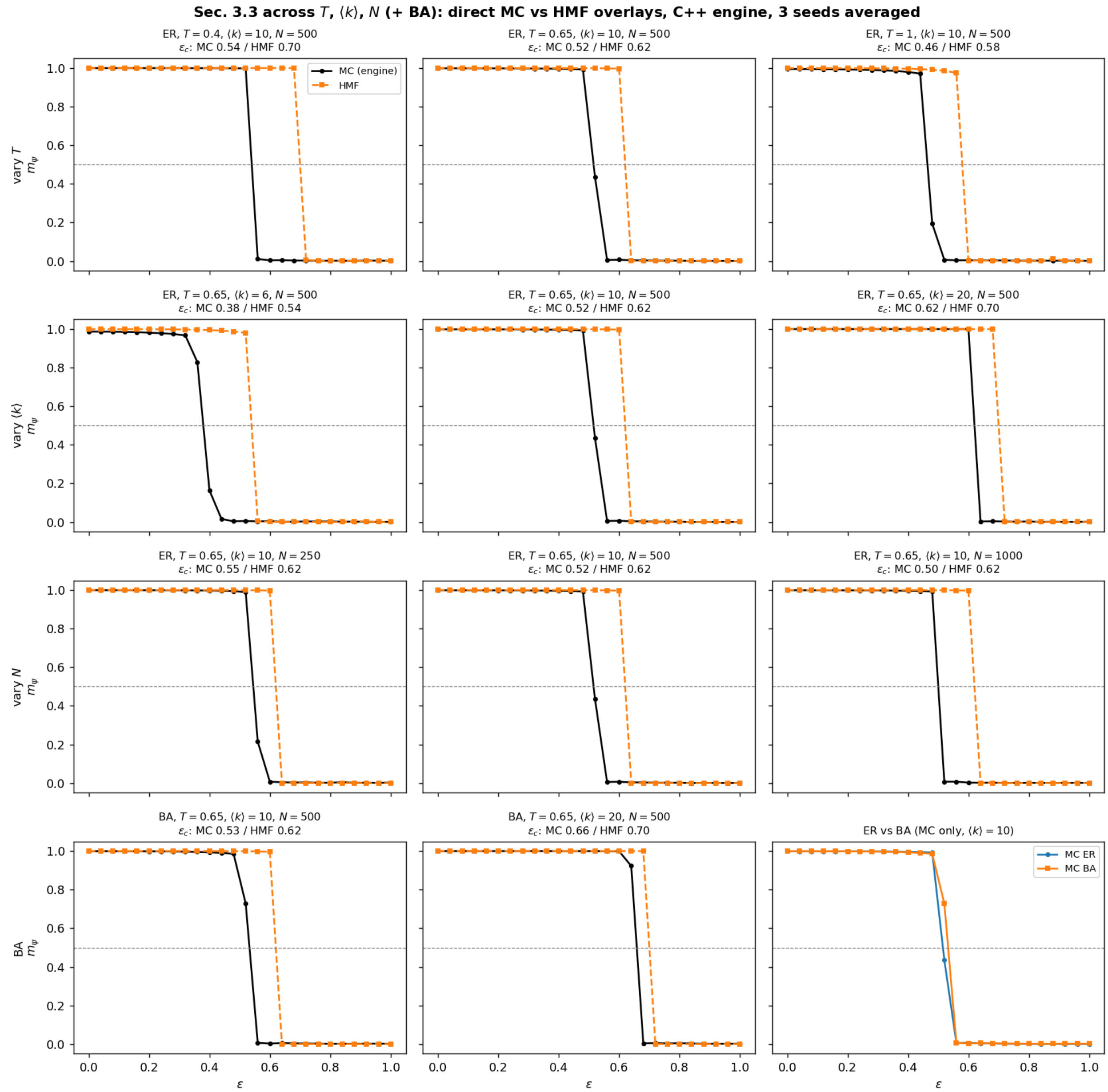


Figure 3: MC (black) vs HMF (orange) overlays across the grid; bottom-right: ER vs BA MC curves coincide.

Verdict vs hypothesis. The $\langle k \rangle$ trend is confirmed: the MC–HMF gap shrinks $0.16 \to 0.08$ from $\langle k \rangle$=6 to 20. In T the gap does *not* grow (0.16 / 0.10 / 0.12 at T=0.4/0.65/1.0): at this degree the HMF tracks the temperature (Sec. 3.2's corrected picture), so the overlays shift together. In N the HMF curve is frozen while the MC crossing walks left ($0.55 \to 0.50$), widening the gap exactly as the $1/N$ drift predicts. BA at matched $\langle k \rangle$ is the same picture as ER (MC ε_c 0.53 vs 0.52; bottom-right panel overlays to line width) — the mean-field error is set by $\langle k \rangle$ and N, not by the degree distribution.

3.4 Robustness: initial composition of the mean field — and a bistable window

What it is. Every mean-field result above starts the map from $(r, p, s)_0 = (0.40, 0.35, 0.25)$. Here the same ε sweep is run from four starting compositions at k=10 and k=20: a near-symmetric start $(0.3334, 0.3333, 0.3333)$, the production default, and two Rock-leaning starts $(0.50, 0.30, 0.20)$ and $(0.90, 0.05, 0.05)$.

Hypothesis — stated before running. The map is deterministic, so only which attractor the start flows to can matter. Deep in either phase the attractor is unique and reached fast: the three biased inits should agree *everywhere*. The near-symmetric start is the predicted exception near ε_c: $(1/3, 1/3, 1/3)$ is an unstable fixed point, and the escape transient can outlast the measurement window where the map slows down.

Parameters — each defined.

- inits — the four starting compositions above; the varied parameter.
- $k \in \{10, 20\}$ — mean degrees fed to the map, bracketing the Sec. 3 operating point.

- ε: 81 points on $[0,1]$ (step 0.0125) — finer than the MC grid because the deterministic map has no sampling noise to hide behind.
- steps = 4000, T=0.65, m_ψ from the last 50% — identical to Secs. 3.1–3.3, so any difference is attributable to the init alone.

How the numbers are obtained. One HMF trajectory (P1 replaced by the map) per (init, k, ε); P3–P4 on the last half; ε_c per curve by P6. Data: `sensitivity/sens_mf_init.csv`.

init $(r,p,s)_0$	ε_c (k=10)	ε_c (k=20)
near-symmetric $(0.3334, 0.3333, 0.3333)$	0.581	0.706
default (production) $(0.4, 0.35, 0.25)$	0.631	0.706
biased $(0.5, 0.3, 0.2)$	0.631	0.706
strongly biased $(0.9, 0.05, 0.05)$	0.706	0.806

Table 6: ε_c depends on the init near the transition: the three biased inits alone span $[0.631, 0.706]$ at k=10 and $[0.706, 0.806]$ at k=20.

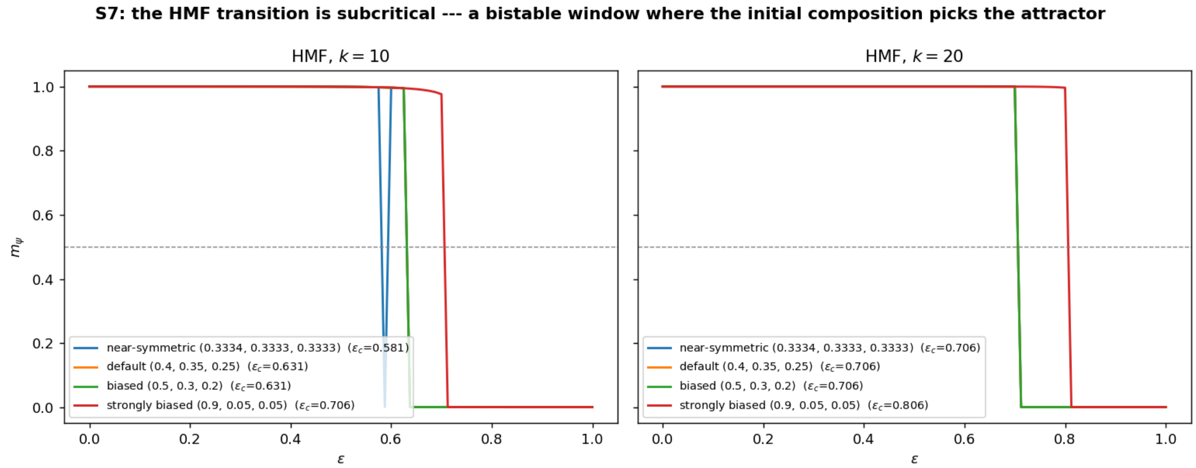


Figure 4: Inside the window the curves are flat at $m_\psi \approx 1$ vs $m_\psi \approx 0$ — two coexisting attractors, not a slow transient.

Verdict vs hypothesis. Half right, and the failure is a discovery. The near-symmetric start misbehaves exactly as predicted (transient dip near the crossing at k=10). But the biased inits do *not* agree near ε_c: within $[0.631, 0.706]$ (k=10) and $[0.706, 0.806]$ (k=20) the consensus fixed point and the limit cycle are *simultaneously stable* and the init picks the attractor. The mean-field transition is therefore subcritical (first-order-like, hysteretic): a mean-field "ε_c" is only defined given an init convention, and the production init sits on the conservative (lower) edge of the window. Secs. 4.2 and 5.1 meet the same first-order signature from independent directions.

4. Phase diagram: $(\langle k\rangle \times \varepsilon)$ MC sweep

What it is. The central result of the study. For each mean degree we build a graph and run full MC across the whole ε range, tiling the $(\langle k\rangle, \varepsilon)$ plane with the measured order parameter to map the order/cycling boundary directly from agent-level dynamics (no mean-field assumption). Run for both ER and BA to isolate whether the boundary is set by average degree or by the shape of $P(k)$. 520 simulations per graph (20 degrees × 26 ε), N=800, fanned across 16 cores in ∼6 s each.

Parameters — each defined.

- N=800 — nodes per graph (fixed, so only connectivity varies along the degree axis).
- $\langle k\rangle = 2, 4, \ldots, 40$ — the 20 average degrees tiling the vertical axis; one graph built per degree (graph seed 1), for ER and for BA.
- ε: 26 points on $[0,1]$ — the horizontal axis: cyclic-dominance strengths (spacing 0.04).
- T=0.65 — Glauber temperature, identical everywhere in the report.
- sweeps = 800, burn-in = 240 — run length per tile and the discarded transient; shorter than elsewhere because 520 runs are needed per panel.
- engine seed = 1 — RNG seed of the dynamics, same for every tile (determinism: rerunning regenerates the CSV byte-for-byte).

- budget — $20 \times 26 = 520$ independent simulations per graph type.

How the numbers are obtained. Every heatmap tile is the m_ψ of one full MC run (P1–P4) at that $(\langle k\rangle, \varepsilon)$ — 520 independent simulations per panel, no averaging or smoothing. Each ε_c table entry applies P6 to the 26-tile column at that degree.

$\langle k\rangle$	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
ε_c (ER)	0.00	0.22	0.38	0.46	0.50	0.54	0.58	0.58	0.59	0.62	0.62	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.70	0.70
ε_c (BA)	0.00	0.22	0.38	0.46	0.50	0.54	0.58	0.58	0.58	0.62	0.62	0.66	0.66	0.66	0.66	0.66	0.70	0.70	0.70	0.70

Table 7: Transition ε_c vs mean degree for ER and BA. The order–cycling boundary bends upward with $\langle k\rangle$; ER and BA are nearly identical $\Rightarrow$ for MC dynamics average degree dominates over the shape of $P(k)$.

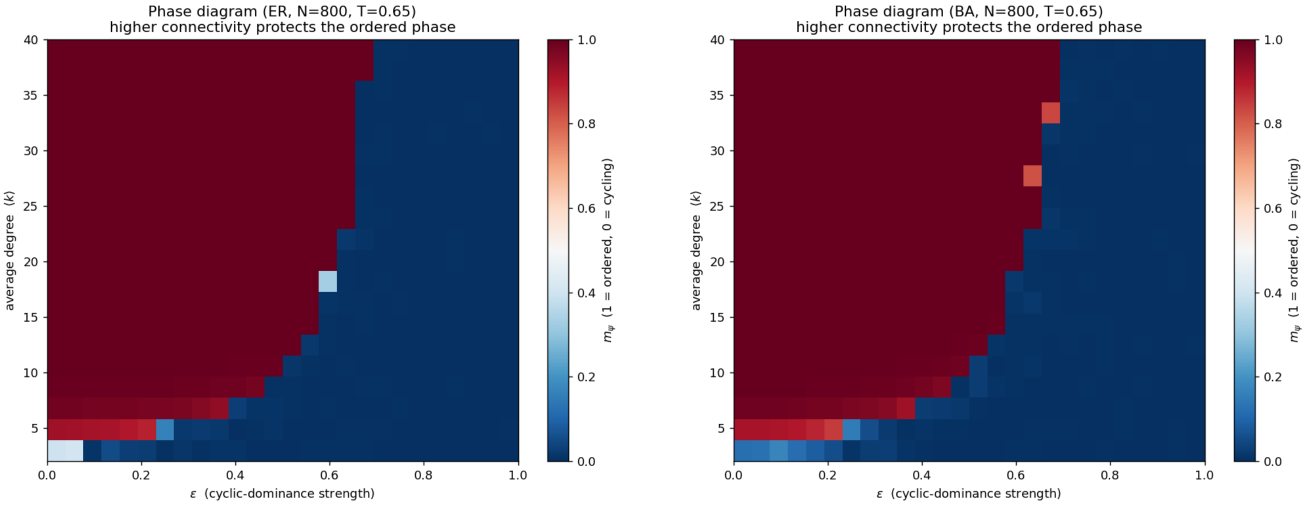


Figure 5: MC phase diagrams: ER (left), BA (right).

Four more diagrams: is the picture an artefact of T=0.65, N=800? The headline diagrams above fix temperature and size. Four additional $(\langle k\rangle \times \varepsilon)$ diagrams (ER) bracket the production point — a cold and a hot one, a small and a large one — each with its extracted boundary over the reference boundary of Sec. 4.1.

Hypothesis — stated before running. The two-phase structure (ordered at small ε, cycling above a $\langle k\rangle$-dependent boundary) is universal across the grid; only the boundary moves. Colder $\to$ boundary up, hotter $\to$ down, with the largest T-shift at small $\langle k\rangle$ (effective noise $\sim T/k$, Sec. 4.2). Smaller $N \to$ boundary up, larger $N \to$ slightly down (the $1/N$ first-order drift, Sec. 5.1); N-shifts smaller than T-shifts at these values.

Parameters — each defined.

- diagrams — $(T, N) \in \{(0.3, 800), (1, 800), (0.65, 300), (0.65, 2000)\}$: two temperatures at the production size, two sizes at the production temperature.
- grid per diagram — 12 degrees $\langle k\rangle \in [2, 40] \times 21$ ε points on $[0, 1]$; ER, graph seed 1, engine seed 1, production sweeps/burn-in.
- reference — the Sec. 4.1 ER boundary (T=0.65, N=800), drawn on every panel and interpolated at the 12 degrees for the shift statistics.

How the numbers are obtained. One P1–P4 run per tile; per-degree ε_c by P6 gives each diagram's boundary; shifts are boundary minus reference at matched degree. Data: `phase_diagram/extra_diagrams.csv` (tiles) and `extra_diagrams_boundary.csv` (boundaries).

diagram	mean shift	shift at $\langle k\rangle$=4	shift at $\langle k\rangle$=40
T=0.3, N=800	+0.024	+0.104	-0.025
T=1, N=800	-0.028	-0.058	+0.025
T=0.65, N=300	+0.040	+0.031	+0.025
T=0.65, N=2000	-0.017	+0.002	-0.025

Table 8: Boundary shift vs the reference (T=0.65, N=800): sign and $\langle k\rangle$-dependence as hypothesised.

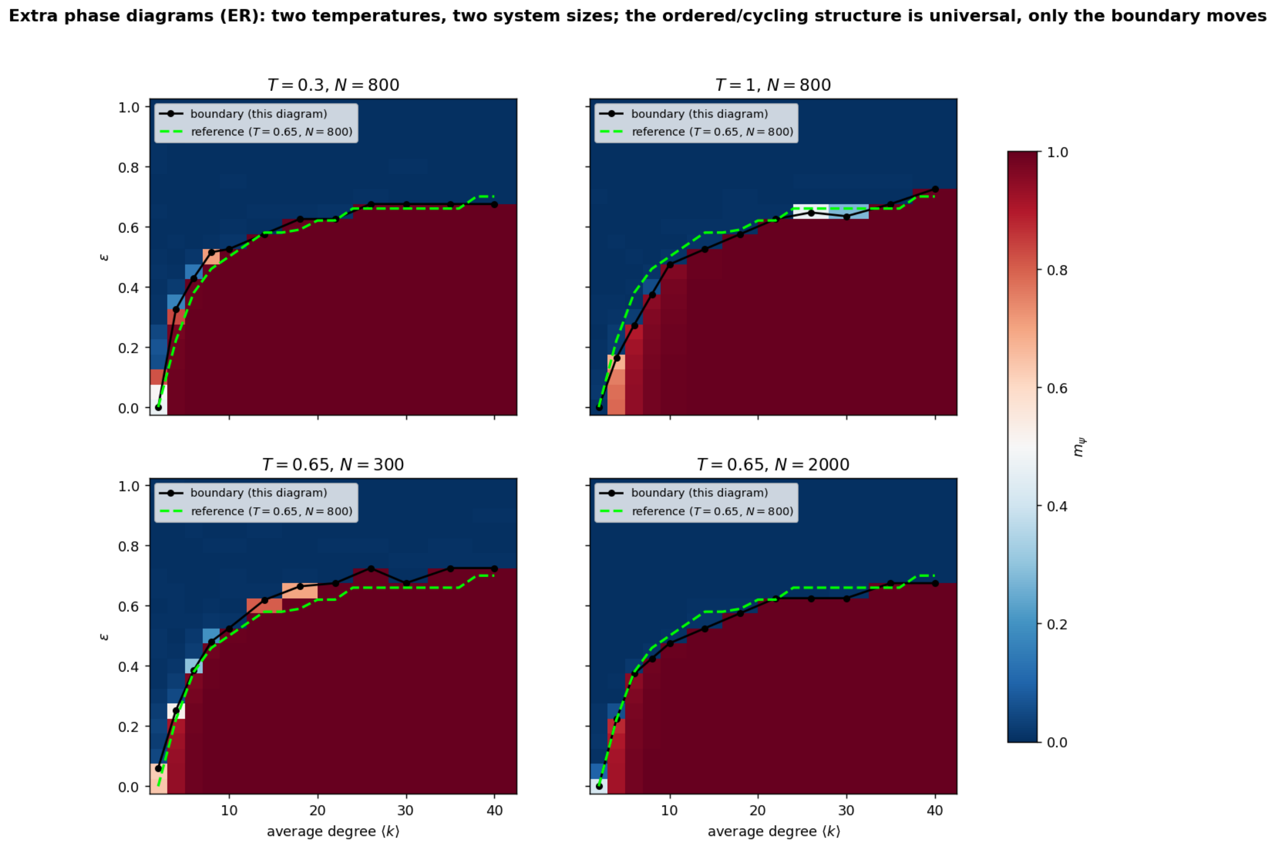


Figure 6: Extra phase diagrams; black: each diagram's boundary, green dashed: the reference. Red = ordered, blue = cycling.

Verdict vs hypothesis. Confirmed on every count. All four diagrams show the same two-phase structure; the boundary moves up when cold (+0.024 mean shift at T=0.3) and down when hot (-0.028 at T=1.0), with the T-shift concentrated at small $\langle k\rangle$ (table) exactly as the T/k effective-noise picture predicts. The size shifts are smaller and signed as the $1/N$ drift requires: +0.040 at N=300, -0.017 at N=2000. The headline diagram is a representative slice of a smooth (T, N) family, not a special point.

4.1 Extracted critical boundary $\varepsilon_c(\langle k\rangle)$

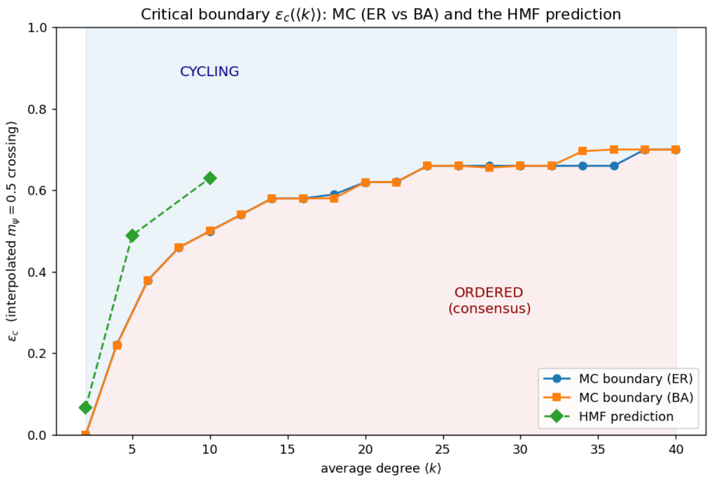
What it is. The boundary pulled out of both heatmaps as a curve (interpolated m_ψ=0.5 crossing per degree), with the HMF prediction from Sec. 3.1 overlaid. One figure carries three claims: the boundary rises with $\langle k\rangle$ (connectivity stabilises order), ER and BA coincide, and the mean field sits above the MC throughout the plotted range (overestimates order; Sec. 4.2 maps where this stops holding).

Parameters — each defined.

- inputs — no new simulations: reads `phase_diagram_{ER,BA}.csv` (the 520-tile MC sweeps of Sec. 4) and `hmf_sweep.csv` (Sec. 3.1).
- ε_c — one value per degree column, by linear interpolation of the m_ψ=0.5 crossing (P6); threshold 0.5 is the midpoint between the two phases.
- clipping $\langle k\rangle \leq 40$ — the HMF sweep extends to k=200; only the range overlapping the MC data is drawn.

quantity	value
$\max_k \lvert\varepsilon_c^{ER} - \varepsilon_c^{BA}\rvert$	**0.040**
boundary range	$0.00 \to 0.70$

The ER and BA boundaries agree within 0.040 at every one of the 20 degrees — the strongest quantitative form of "average degree matters, not $P(k)$". Each boundary point is P6 applied to one degree column of the Sec. 4 heatmaps. Data: `critical_boundary.csv`.



4.2 Robustness: temperature × degree

What it is. The whole phase diagram is taken at T=0.65. Here ε_c is measured on a $T \times \langle k\rangle$ grid (MC, seed-averaged) with the HMF prediction computed on the same grid, to establish how the headline "$\varepsilon_c(\langle k\rangle)$" depends on the one thermodynamic parameter it froze.

Hypothesis — stated before running. T is a genuine control parameter: the Glauber acceptance compares the payoff gain to T, so more noise weakens the ferromagnetic pinning and ε_c should fall monotonically with T at every $\langle k\rangle$. But the payoff a node feels scales with its degree ($\Delta U \sim k$), so the *effective* noise is $\sim T/k$ and the T-dependence should flatten as $\langle k\rangle$ grows. HMF should reproduce the trend while overestimating ε_c everywhere. The two-phase structure itself should survive at every (T, k).

Parameters — each defined.

- $T \in \{0.3, 0.5, 0.65, 0.8, 1, 1.3\}$ — Glauber temperatures bracketing the production value 0.65 on both sides; the varied thermodynamic knob.
- $\langle k\rangle \in \{8, 20, 40\}$ — mean degrees; varied jointly with T to expose the interaction.
- graphs — ER, N=500, 4 graph seeds (P5); engine seed = graph seed.
- ε: 21 points on $[0, 1]$ (step 0.05), T passed to the engine; 1500 sweeps, burn-in 30% — otherwise the production protocol.
- HMF — same (T, k) grid, 4000 map steps, ε_c from a fine 0.01 ε grid (the map is deterministic, so its crossing need not be quantised by the MC grid).

How the numbers are obtained. P1–P4 per $(T, k, \varepsilon, \text{seed})$, ε_c by P6 per seed then averaged (P5), std across seeds as the uncertainty; HMF column by the map + P6. Data: `sensitivity/sens_temperature.csv`.

	$\langle k\rangle$=8		$\langle k\rangle$=20		$\langle k\rangle$=40	
T	MC ε_c	HMF	MC ε_c	HMF	MC ε_c	HMF
0.3	0.521±0.006	0.695	0.648±0.023	0.655	0.725±0.000	0.665
0.5	0.478±0.004	0.645	0.625±0.000	0.685	0.725±0.000	0.655
0.65	0.475±0.000	0.605	0.625±0.000	0.705	0.714±0.020	0.655
0.8	0.428±0.004	0.565	0.625±0.000	0.685	0.713±0.012	0.685
1	0.390±0.017	0.505	0.616±0.005	0.635	0.705±0.015	0.685
1.3	0.324±0.001	0.425	0.575±0.000	0.635	0.698±0.023	0.705

Table 9: MC ε_c falls monotonically with T at every degree, and the fall flattens with $\langle k\rangle$: total drop over $T \in [0.3, 1.3]$ 0.197 at $\langle k\rangle$=8, 0.073 at $\langle k\rangle$=20, 0.027 at $\langle k\rangle$=40.

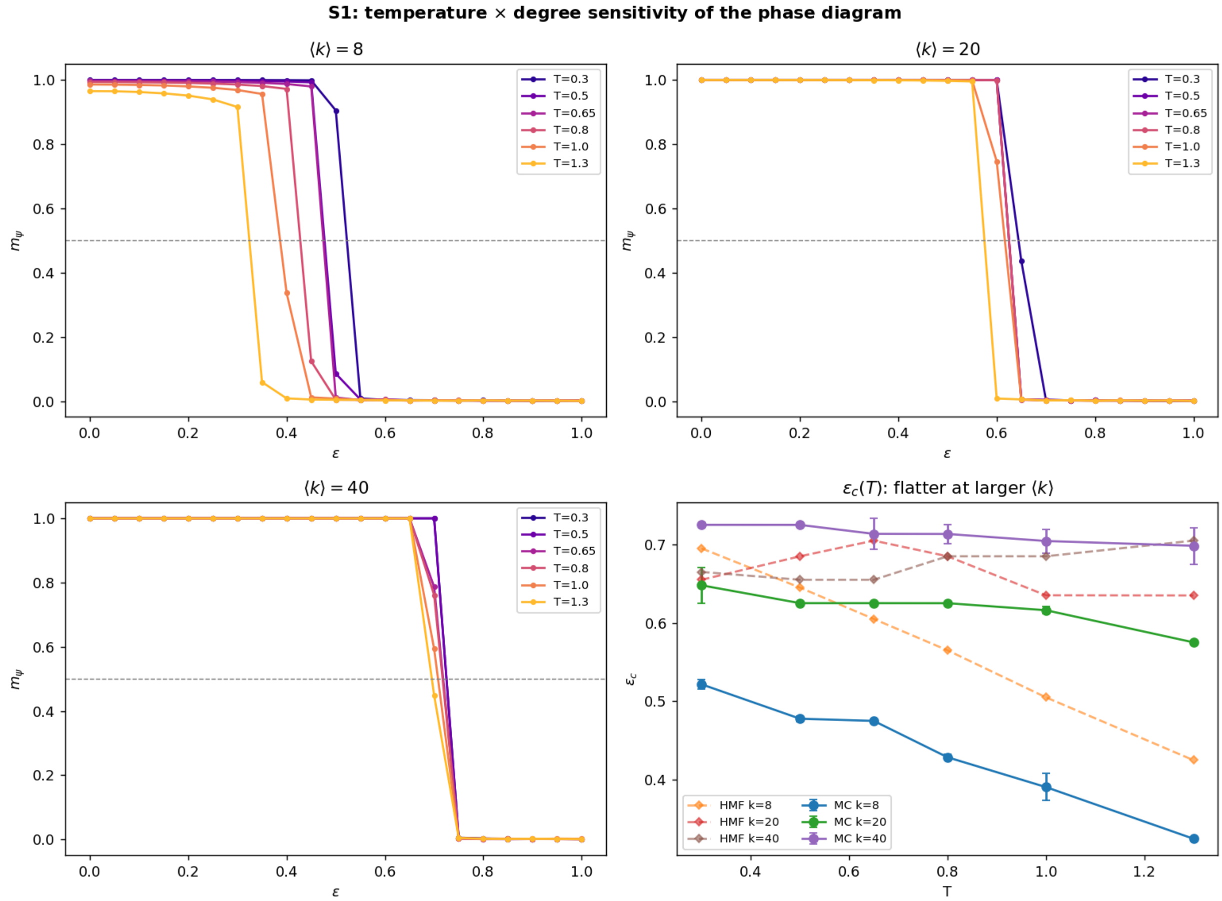


Figure 7: Transition curves per degree (colour = T) and the extracted $\varepsilon_c(T)$ surface (bottom right).

Verdict vs hypothesis. MC part confirmed on both counts: monotone decrease at every degree, and the $T \times k$ interaction is exactly as predicted — the drop shrinks $0.197 \to 0.027$ from $\langle k \rangle$=8 to 40, so at T=0.65 the system is deep in the k-dominated regime, which is what licenses the report's "stability is a function of $\langle k \rangle$ alone". The HMF half is *refuted* at high degree: the overestimate holds at $\langle k \rangle$=8 (gap +0.10 to +0.17) but shrinks and *reverses* by $\langle k \rangle$=40 (gap −0.07 to +0.01), wiggling non-monotonically in T. Explanation via Sec. 3.4: for $kU \gg T$ the mean-field rates saturate to step functions, so the map's ε_c plateaus near 0.7 while the MC boundary keeps rising; and inside the bistable window a single-init ε_c measures a basin boundary, not a bifurcation point — hence the wiggle. The "HMF sits above MC" claim of Sec. 4.1 is drawn for $\langle k \rangle \leq 10$, where it is correct; this experiment marks its domain of validity.

4.3 Robustness: graph seed vs MC seed

What it is. Are single-realisation results representative? 5 graph seeds × 5 engine seeds (= 25 combinations) each run the full ε sweep at the baseline point (ER, N=500, $\langle k \rangle$=20).

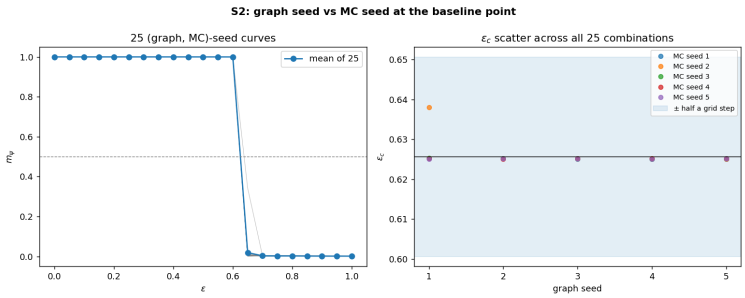
Hypothesis — stated before running. Both seeds are nuisance parameters at this size: ε_c should scatter well under one grid step (0.05) across all combinations; m_ψ scatter should peak at the crossing (critical fluctuations) and vanish deep in either phase; and the graph-seed and MC-seed contributions should be *comparable*, because an ER graph at $\langle k \rangle$=20 is locally homogeneous — no realisation owns structure (hubs, modules) that could shift the transition. If graph scatter dominated, the single-graph sweeps of Secs. 2–4 would not be representative.

Parameters — each defined.

- graph seeds {1..5} — independent ER realisations (quenched disorder).
- MC seeds {1..5} — independent dynamics streams on each graph (thermal noise).
- baseline system — ER, N=500, $\langle k \rangle$=20, T=0.65, 1500 sweeps, burn-in 30%, ε: 21 points on $[0, 1]$.

How the numbers are obtained. P1–P4 and P6 per combination (no averaging — the scatter *is* the measurement); the two seed types are separated by averaging ε_c over one index and taking the std over the other. Data: `sensitivity/sens_seeds.csv`.

quantity	value
ε_c mean	0.6257
total std (25 combos)	0.0025
range	0.0130
graph-seed std	0.0010
MC-seed std	0.0010
m_ψ scatter peak	at $\varepsilon = 0.65$



Verdict vs hypothesis. Confirmed on all three counts: total ε_c std 0.0025 is ∼20× below the grid step; the scatter of m_ψ peaks exactly at the crossing ($\varepsilon = 0.65$, cf. $\varepsilon_c = 0.63$) and is zero at ε=0; and the graph/MC decomposition is 0.0010 vs 0.0010 — indistinguishable, as local homogeneity predicts. Seed averaging in this report buys cosmetic smoothness, not correctness; single-realisation sweeps are representative.

4.4 Robustness: ε-grid resolution and the ε_c estimator

What it is. One fine sweep (step 0.0125, 4 seeds averaged, baseline system) is *subsampled* to steps 0.025, 0.05 and 0.10, so every resolution sees identical simulation data and only the grid handed to the estimator changes. Both the project estimator (P6, interpolated) and the naive first-point-below convention are applied to each grid.

Hypothesis — stated before running. The interpolated estimator makes grid step a nuisance parameter: $m(\varepsilon)$ is near-linear across one step at the crossing, so the ε_c shift should stay well below half a step even at 0.10, converging as the step shrinks. The naive estimator should be biased by up to a full step, and always upward (it can only round toward larger ε).

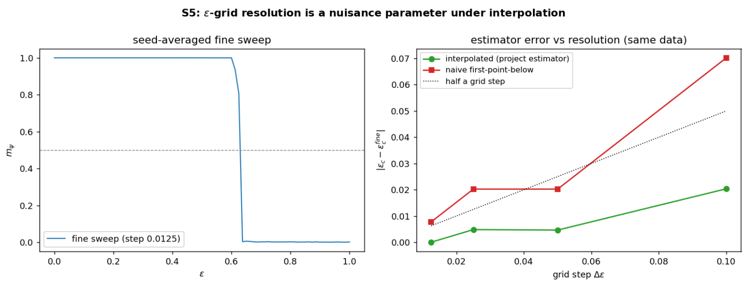
Parameters — each defined.

- base sweep — 81 ε points on $[0, 1]$ (step 0.0125), ER, N=500, $\langle k\rangle$=20, 4 graph seeds averaged (P5).
- grid steps $\in \{0.0125, 0.025, 0.05, 0.1\}$ — exact subsampling of the same curve; the varied parameter.
- estimators — P6 (interpolated crossing) vs the first grid point with $m_\psi < 0.5$ (the project's old convention, kept as the control).

How the numbers are obtained. No new simulations beyond the base sweep: subsample, apply both estimators, compare to the finest-grid value. Data: `sensitivity/sens_grid.csv`.

step	interp. shift	naive shift
0.0125	+0.0000	+0.0077
0.025	+0.0048	+0.0202
0.05	-0.0046	+0.0202
0.1	+0.0203	+0.0702

Shifts relative to the finest-grid interpolated value (0.6298).



Verdict vs hypothesis. Confirmed. At the production step 0.05 the interpolated estimator is off by -0.0046 (a tenth of a step) while the naive one is off by +0.0202; at step 0.10 the naive bias reaches +0.0702 — order a full step and always positive, as predicted. Interpolation buys ∼4× effective resolution; the project-wide convention (P6) is validated, and every ε_c in this report inherits a grid uncertainty of at most ∼0.005.

5. Dynamics and finite-size scaling

What it is. Three views that confirm the transition is a real phase transition rather than a finite-size or dynamical artefact. **FSS:** rerun the MC transition curve at growing system size N and watch it sharpen and its midpoint converge. **Ternary portrait:** plot the trajectory in the (r, p, s) composition simplex — a fixed corner means consensus, a closed orbit means the endless RPS chase. **Stability:** classify the mean-field fixed points (consensus vs limit cycle) on either side of ε_c.

Parameters — each defined.

- **FSS** (left panel)
  - graph — ER at fixed $\langle k\rangle$=10, so only size varies.

– $N \in \{200, 500, 1000, 2000\}$ — the system sizes compared: the control variable of finite-size scaling (graph seed 1 for each).
– ε: 21 points on $[0.35, 0.75]$ — grid deliberately zoomed onto the transition region for resolution where the curves are steep.
– engine — T=0.65, 1200 sweeps (burn-in 360 discarded), seed 1.

- **Ternary portrait** (middle panel)
  – graph — ER, N=500, $\langle k\rangle$=10, graph seed 1.
  – dynamics — pure-Python MC, 1200 sweeps, MC seed 3; the raw per-sweep (r, p, s) trajectory is plotted, so no burn-in or averaging.
  – $\varepsilon \in \{0.2, 0.95\}$ — one value per phase: consensus corner vs cycling orbit.
- **Stability** (right panel)
  – model — HMF map at k=10, T=0.65, 600 steps (enough to see decay or growth of the perturbation).
  – init $(1/3+\delta,\ 1/3,\ 1/3-\delta)$, δ=0.02 — a small kick off the symmetric fixed point, whose fate classifies the fixed point's stability.
  – $\varepsilon \in \{0.2, 0.5, 0.7, 0.95\}$ — values straddling ε_c, to show the switch from stable consensus to limit cycle.

How the numbers are obtained. FSS: one engine run (P1–P4) per (N, ε); ε_c row by P6 per N-column; max|slope| is the discrete steepness $\max_j |m_{j+1}-m_j|/(\varepsilon_{j+1}-\varepsilon_j)$ of that column — it must grow with N if the transition is sharpening. Ternary: the raw per-sweep (r, p, s) series from P2 plotted directly in the simplex (no averaging), so consensus shows as a point and cycling as an orbit. Stability: the HMF map is started a distance δ from the symmetric point and the trajectory is classified by whether the perturbation decays or grows into a limit cycle.

N	200	500	1000	2000
ε_c	0.556	0.520	0.500	0.499
max|slope|	33	49	49	44

Table 10: Finite-size scaling (ER, $\langle k\rangle$=10): the transition sharpens and ε_c converges toward ≈0.50 as N grows 200→2000.

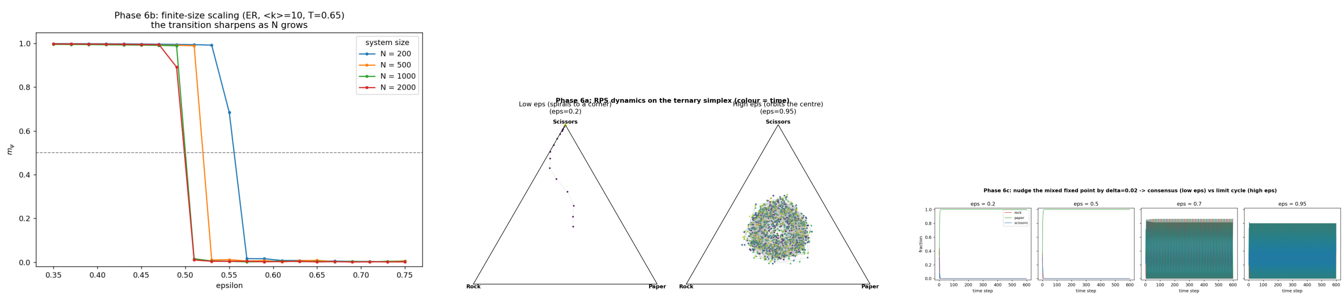


Figure 8: Left: FSS. Middle: ternary phase portrait (corner consensus below ε_c, limit cycle above). Right: fixed-point stability.

5.1 Robustness: system size at $\langle k\rangle$=20

What it is. The FSS panel above works at $\langle k\rangle$=10; this extends it to the $\langle k\rangle$=20 operating point used by the robustness suite, with full ε sweeps at six sizes and a fine grid (step 0.0125) across the transition.

Hypothesis — stated before running. ε_c is a property of the local neighbourhood ($\langle k\rangle$), not of N: the crossing should drift only weakly and settle by $N \sim 1000$. Three finite-size effects *should* scale away: the transition width shrinks, seed scatter shrinks, and cycling-phase m_ψ falls as $1/\sqrt{N}$ (finite-N cycling amplitude is a fluctuation effect).

Parameters — each defined.

- $N \in \{125, 250, 500, 1000, 2000, 4000\}$ — five octaves of system size; the varied parameter (ER, $\langle k\rangle$=20 fixed).
- seeds — 4 graph seeds per size (P5); engine seed = graph seed.
- ε grids — 21 points on $[0, 1]$ (curves, cycling metrics) + 29 points on $[0.45, 0.80]$ at step 0.0125 (ε_c, width): the production grid quantises the N-drift because the transition is sharper than one 0.05 step.
- engine — T=0.65, 1500 sweeps, burn-in 30% (production protocol).
- width — distance between the m_ψ=0.75 and 0.25 crossings (interpolated), a resolution-independent sharpness measure.

How the numbers are obtained. P1–P4 per $(N, \varepsilon, \text{seed})$; ε_c by P6 per seed on the fine grid, then mean±std (P5); cycling metrics from the coarse grid ($\varepsilon \geq 0.8$). Data: `sensitivity/sens_size.csv`.

N	ε_c	width	m_ψ (cycling)	seed scatter (cycling)
125	0.700±0.013	0.021	0.0040	0.0023
250	0.665±0.005	0.013	0.0027	0.0015
500	0.629±0.002	0.008	0.0019	0.0011
1000	0.610±0.005	0.010	0.0012	0.0005
2000	0.594±0.000	0.006	0.0011	0.0005
4000	0.587±0.006	0.012	0.0006	0.0004

Table 11: ε_c drifts *down* smoothly with N; the gap to 0.58 halves per doubling of N ($\sim 1/N$), extrapolating (Richardson, two largest sizes) to $\varepsilon_c(\infty) \approx 0.580$ at $\langle k\rangle$=20.

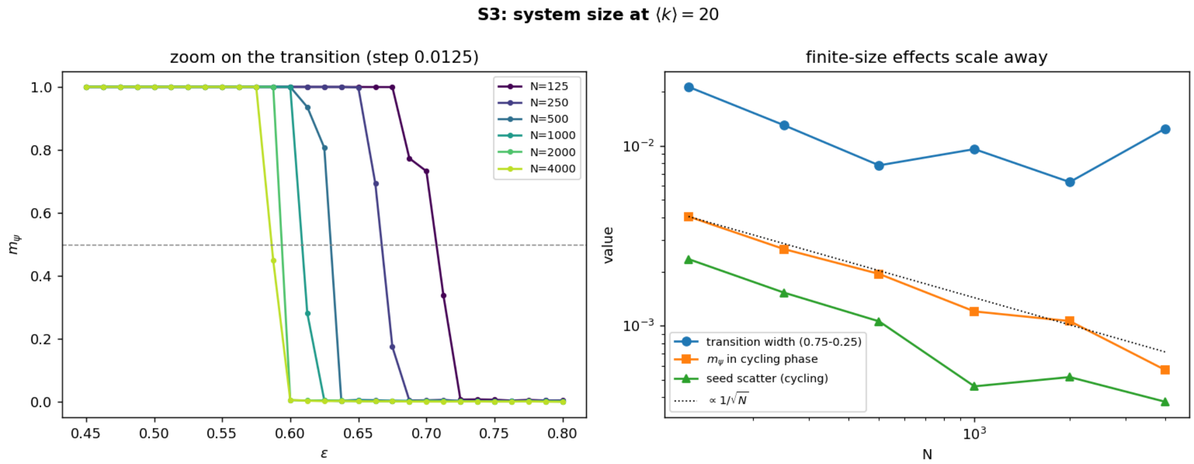


Figure 9: Left: fine-grid transition curves. Right: cycling-phase metrics vs the $1/\sqrt{N}$ guide (log–log).

Verdict vs hypothesis. Half confirmed, half corrected by the data. Confirmed: cycling-phase m_ψ follows $1/\sqrt{N}$ over five octaves ($0.0040 \to 0.0006 \approx 1/\sqrt{32}$ of the start), seed scatter shrinks alongside, and the transition sharpens (the width metric bottoms out at the 4-seed noise floor $\sim$0.01). Corrected: "settles by $N \sim 1000$" is wrong — the crossing keeps sliding, 0.700 $\to$ 0.587, as $\sim 1/N$. A $1/N$ shift of the pseudo-transition point is the standard finite-size scaling of a *first-order* transition — the same subcriticality Sec. 3.4 found in the mean field, now visible in the agent-level MC. Practical reading: quoted ε_c values are N-finite estimates (at N=500, high by ≈ 0.05 at this degree); every *comparison* in this report (ER vs BA, damaged vs pristine, MC vs MF) is made at matched N, so the conclusions are unaffected.

5.2 Robustness: simulation length

What it is. Does 1500 sweeps suffice? The baseline sweep is repeated at six run lengths (burn-in held at 30% of each), with the 6000-sweep curve as the reference.

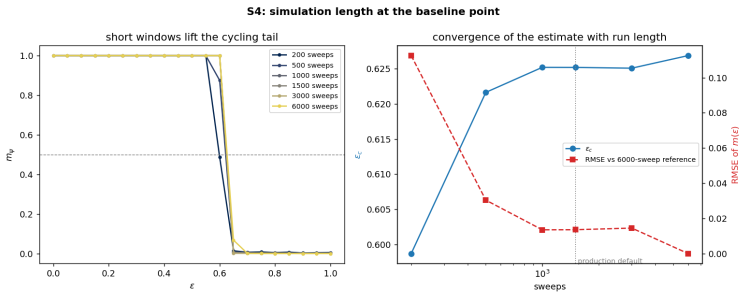
Hypothesis — stated before running. Two opposite finite-*time* biases, one per phase. Cycling side: $m_\psi = |\overline{\psi}|$ vanishes only when the window covers many rotations of $\psi(t)$, so a short window biases it *up*. Ordered side: a short run may not finish ordering from the random start, biasing m_ψ *down*. The cycling-side bias should push the m_ψ=0.5 crossing right, so ε_c is *over*estimated at small sweeps, converging from above by $\sim$1500. RMSE against the reference should fall monotonically.

Parameters — each defined.

- sweeps $\in \{200, 500, 1000, 1500, 3000, 6000\}$ — run lengths; the varied parameter. One sweep = N attempted updates.
- burn-in = 30% of each run — scales with the run so the discarded transient fraction is constant (the production convention).
- baseline system — ER, N=500, $\langle k\rangle$=20, T=0.65, 4 graph seeds (P5), ε: 21 points on $[0, 1]$.
- reference — the 6000-sweep seed-averaged curve; RMSE is computed against it over the whole grid.

How the numbers are obtained. P1–P4 per (sweeps, ε, seed) with burn-in 0.3×sweeps, seed-averaged (P5); ε_c by P6 per run length; RMSE vs the 6000-sweep curve. Data: `sensitivity/sens_equilibration.csv`.

sweeps	ε_c	RMSE	m_ψ (cyc.)
200	0.599	0.113	0.0055
500	0.622	0.030	0.0037
1000	0.625	0.014	0.0025
1500 (default)	0.625	0.014	0.0019
3000	0.625	0.015	0.0011
6000	0.627	0.000	0.0008



Verdict vs hypothesis. Mechanisms right, sign wrong, default validated. Both predicted biases exist, but their sizes were misjudged: the cycling-side up-bias is real yet tiny (0.0055 at 200 sweeps vs 0.0008 at 6000 — nowhere near the 0.5 threshold), while the ordered-side down-bias is large: at 200 sweeps the run cannot finish ordering near the transition, m_ψ dips below 0.5 early, and ε_c comes out *under*estimated (0.599 vs 0.627). The crossing sits on the steep ordered flank, so that bias owns the sign. Convergence is complete by 1000 sweeps (ε_c flat, RMSE at the 0.014 seed-noise floor): the production 1500 carries a $\sim 2\times$ safety margin.

6. Perturbation experiments

These studies build on the phase diagram: having mapped the clean transition, we now stress the ordered phase with targeted perturbations — stubborn agents, structural targeting, competition, and quenched network damage — to ask how robust consensus is and how a cyclic (RPS) system responds to interference. Each is run in both the ordering phase (ε=0.3) and the cycling phase (ε=0.9), averaged over many graph seeds.

6.1 Single Rock-zealot faction (ER, $\langle k\rangle$=10, 12 seeds)

What it is. A fraction z of nodes are turned into *zealots* permanently locked to Rock (they influence neighbours but never update). We grow z and track two things: *conversion* (what fraction of the remaining *free* nodes end up playing Rock) and the global order m_ψ. The classic question of whether a committed minority can drive consensus — but asked in a cyclic system, where pushing Rock also feeds Rock's predator.

Parameters — each defined.

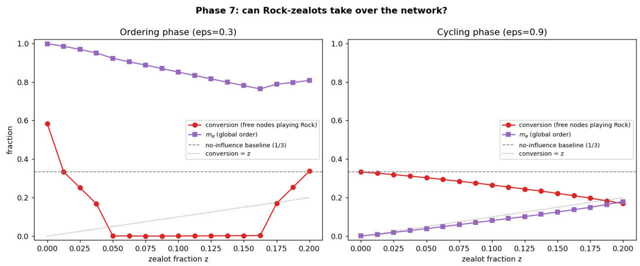
- graphs — ER, N=800 nodes, $\langle k\rangle$=10 (average degree); 12 independent realisations (seeds 1–12), every plotted point is the 12-seed average (P5).
- zealot strategy = Rock — the strategy the committed minority is locked to (by symmetry the choice is arbitrary).
- placement = random — the $\lfloor zN\rceil$ zealot nodes are drawn uniformly from the network.
- z: 17 points on $[0, 0.20]$ — zealot fraction, the control variable (up to one node in five committed).
- engine — T=0.65 (Glauber temperature), 1500 sweeps, burn-in 450 (discarded).
- ε=0.3 / 0.9 — one value inside each phase: the ordering panel and the cycling panel of the figure.

How the numbers are obtained. For each $(z, \varepsilon, \text{seed})$: one engine run in which the $\lfloor zN\rceil$ zealot nodes are drawn uniformly, locked to Rock, and skipped by the update loop (P1). Conversion is recomputed every post-burn-in sweep as (free nodes playing Rock)/(free nodes) — zealots excluded from numerator and denominator — then time-averaged like m_ψ (P4) and averaged over the 12 graphs (P5). The table shows every 4th point of the 17-point z grid.

z	$\mathrm{conv}_{\mathrm{ord}}$	$m_{\psi\,\mathrm{ord}}$	$\mathrm{conv}_{\mathrm{cyc}}$	$m_{\psi\,\mathrm{cyc}}$
0.00	0.58	1.00	0.33	0.00
0.05	0.00	0.92	0.30	0.04
0.10	0.00	0.85	0.27	0.08
0.15	0.00	0.78	0.22	0.13
0.20	0.34	0.81	0.17	0.18

conv = fraction of *free* nodes playing Rock. Ordering phase: Rock-conversion collapses to ∼0 by z≈0.05 — the free network flips to **Paper** (Rock's predator). Cycling phase: only weak induced order, strategy cannot be pinned.

Conclusion. A committed minority cannot spread its own strategy in a cyclic system — every zealot acts as food for its predator, so pushing Rock elects Paper. And the two phases fail oppositely: the ordered phase is *compositionally fragile* (5% zealots dictate the winner: free-Rock conversion 0.00 at z=0.05) while the cycling phase is nearly immune (m_ψ only 0.18 even at z=0.20).



The same experiment across T, $\langle k \rangle$ and N. Is the backfire a property of the operating point or of the model? The protocol above is repeated with one axis varied at a time around (ER, N=800, $\langle k \rangle$=10, T=0.65).

Hypothesis — stated before running. The backfire is generic: in every cell, small-z Rock-zealots should hand the ordering-phase network to Paper, never to Rock. T controls the *sharpness*: hotter systems pin the beater-consensus less strongly and need more zealots before the large-z re-pinning on Rock appears. $\langle k \rangle$ strengthens local majority pressure (effective noise $\sim T/k$), so denser graphs backfire more cleanly. N is a null axis: conversion and m_ψ are intensive, so the curves should coincide within seed noise, only smoother at larger N.

Parameters — each defined.

- $T \in \{0.4, 0.65, 1.0\}$, $\langle k \rangle \in \{6, 10, 20\}$, $N \in \{400, 800, 1600\}$ — one axis at a time (7 distinct cells).
- z: 9 fractions on $[0, 0.2]$; zealots Rock, random placement; 8 graph seeds per point (P5); both phases (ε=0.3 and 0.9).
- observables — conversion (Sec. 0.1), m_ψ, and ρ_{Paper} (the beater's global fraction), per cell and z.

How the numbers are obtained. P1–P4 per (cell, phase, z, seed) with the zealot machinery of Sec. 6.1, then P5; the figure shows mean±std over seeds. Data: `zealots/zealots_grid.csv`.

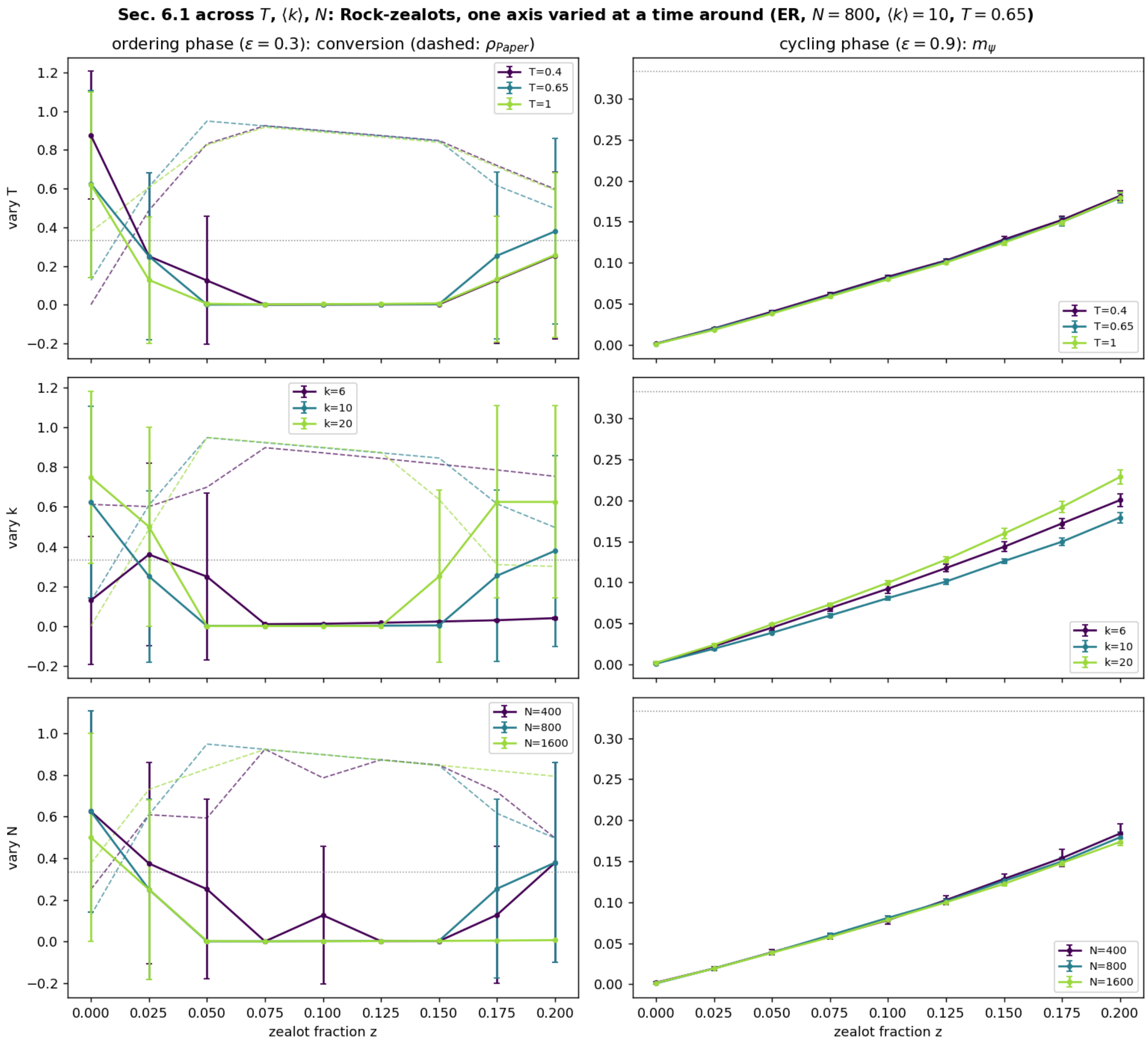


Figure 10: Rows: vary T / $\langle k \rangle$ / N. Left: ordering-phase conversion (dashed: ρ_{Paper}). Right: cycling-phase m_ψ.

Verdict vs hypothesis. The backfire is generic: at z=0.05 the beater's population ρ_{Paper} reaches 0.83/0.95/0.82 for T=0.4/0.65/1.0 and 0.70/0.95 for $\langle k \rangle$=6/20 — Rock-zealots elect Paper in every cell, never Rock. The T prediction holds: the large-z re-pinning on Rock weakens with heat (conversion at z=0.2: 0.25/0.38/0.26), and the cycling phase stays immune everywhere (right column). N is a null axis for the *deterministic* part of the response: wherever no seed flips to Rock at any size, the conversion gap across N is ≤ 0.002. What N does change is the *probability* of the stochastic large-z flips onto the zealots' own strategy: the flipped fraction of seeds at z=0.2 falls 0.38 $\rightarrow$ 0.38 $\rightarrow$ 0.01 for N=400/800/1600 — flipping the whole free network is a collective fluctuation that becomes rarer with size, so larger systems backfire *more* reliably, refining rather than breaking the null-axis prediction.

The zealot story as a time signal. Everything above is time-averaged; here four runs are shown *per sweep* — who led at the start, what the zealots changed, when the lead flipped, and where it ended. One ER graph (N=800, $\langle k \rangle$=10, seed 3), per-sweep global (r, p, s) recorded by the engine (no RNG cost, so these are exactly the runs the summary numbers come from).

Parameters — each defined.

- scenarios — (ε, z): $(0.3, 0)$ clean ordering; $(0.3, 0.05)$ the backfire; $(0.3, 0.20)$ a large faction; $(0.9, 0.10)$ cycling. Zealots Rock, random placement.
- recording — every sweep $t = 0 \ldots 1500$ including burn-in (the story needs the transient the other sections discard); log time axis because consensus on a dense random graph forms within $\sim$10 sweeps.

How the numbers are obtained. One engine run per scenario with `--timeseries`; the story table lists the t=0 composition (P2 at t=0), the final winner (largest tail-averaged fraction), the sweep it took the lead for good, and the sweep it crossed 50%. Data: `zealots/timeseries.csv` (signals), `timeseries_story.csv` (story).

scenario	start (r, p, s)	initial leader	winner	majority at	final (r, p, s)
clean ordering (ε=0.3, z=0)	(0.32, 0.34, 0.34)	Paper	Paper	sweep 6	(0.00, 1.00, 0.00)
backfire (ε=0.3, z=0.05)	(0.35, 0.33, 0.32)	Rock	Paper	sweep 4	(0.05, 0.95, 0.00)
large faction (ε=0.3, z=0.2)	(0.45, 0.29, 0.26)	Rock	Paper	sweep 8	(0.20, 0.79, 0.00)
cycling (ε=0.9, z=0.1)	(0.39, 0.32, 0.29)	Rock	none (cycling)	never	(0.34, 0.38, 0.28)

Table 12: The story table, computed from the recorded signals.

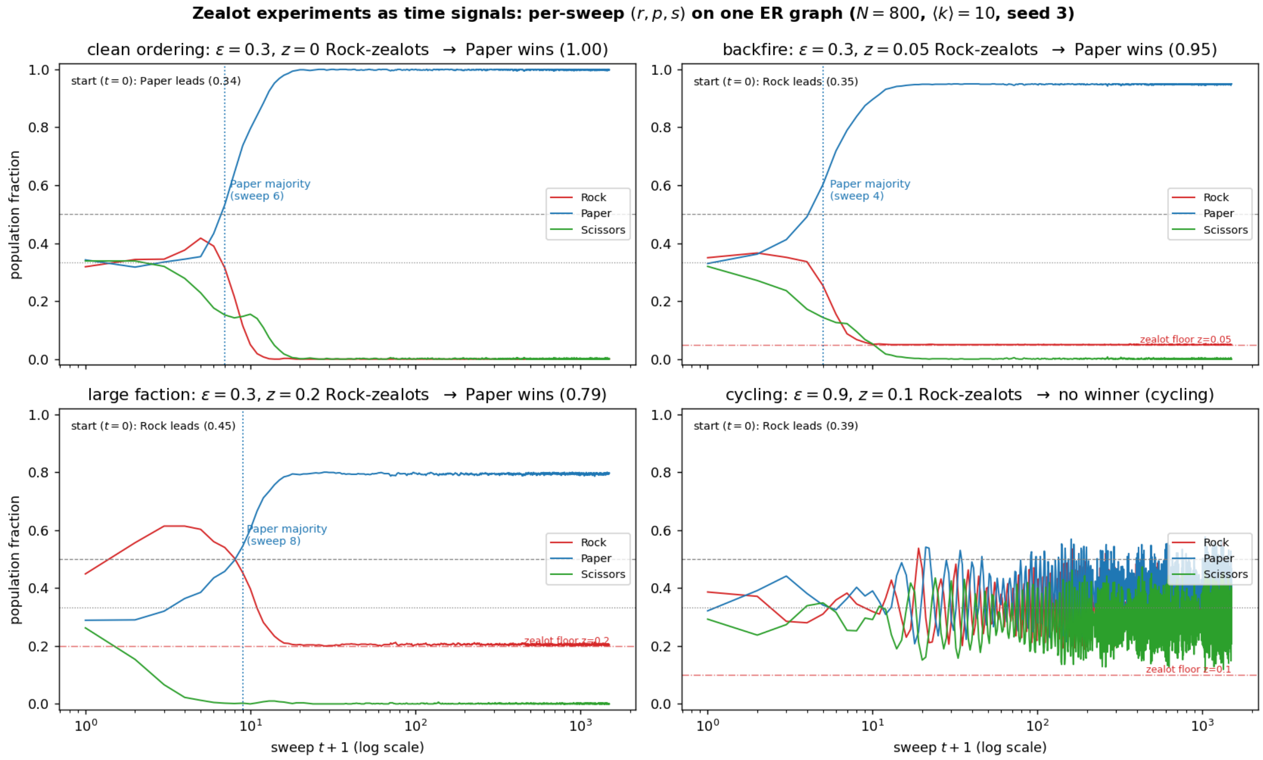


Figure 11: Per-sweep population fractions (log time). Dash-dotted red line: the zealot floor z below which Rock cannot fall.

Reading the signals. *Clean ordering*: all three start near 1/3 (Paper marginally ahead at 0.34); whoever leads early snowballs, and by sweep 6 it is a majority — the winner is decided by the random start. *Backfire*: the zealots make Rock the initial leader (0.35), yet Paper — Rock's predator — overtakes immediately, is a majority by sweep 4, and Rock is eaten down to its zealot floor z=0.05 (final r=0.05): planting zealots *selected the winner for the enemy*. *Large faction*: Rock first *grows* (to 0.61, feeding on Scissors) before Paper catches up at sweep 8 and pins it at the floor z=0.2 — the rise-then-collapse is the cyclic mechanism in real time. *Cycling*: no winner ever; the oscillations grow into the noisy steady chase and the zealots change nothing — exactly why the time-*averaged* m_ψ of Sec. 6.1 is the right order parameter.

6.2 Hub vs random placement (BA, 15 seeds)

What it is. The same Rock-zealots, but now we compare *where* they sit on a scale-free BA network: spread at random versus concentrated on the highest-degree *hubs*. Since hubs touch far more neighbours, this tests how much raw structural targeting amplifies a minority's reach — and whether that extra reach lets it dictate *which* strategy wins, or merely *whether* the network orders at all.

Parameters — each defined.

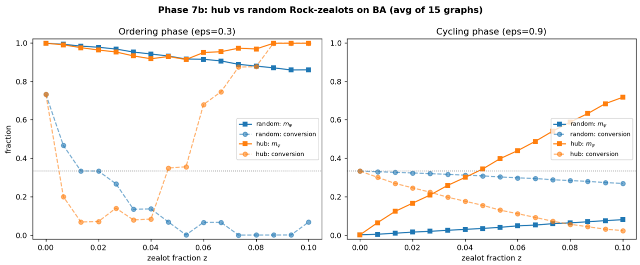
- graphs — BA (scale-free, so hubs exist), N=800, $\langle k \rangle$=10; 15 realisations (seeds 1–15), curves are the 15-seed average (P5).
- zealot strategy = Rock — as in Sec. 6.1.
- placement $\in$ {random, hub} — the variable under test: *random* draws the zealot set uniformly, *hub* locks the $\lfloor zN \rceil$ highest-degree nodes; same budget, different position.
- z: 16 points on $[0, 0.10]$ — zealot fraction; capped at 10% because hub placement saturates early.
- engine — T=0.65, 1500 sweeps, burn-in 450.
- $\varepsilon \in \{0.3, 0.9\}$ — ordering-phase and cycling-phase panels.

How the numbers are obtained. Identical pipeline to Sec. 6.1 (P1–P5 with locked zealot nodes), except the placement rule: *random* draws the zealot set uniformly, *hub* sorts nodes by degree and locks the top $\lfloor zN \rceil$. The quoted amplification is the ratio of the two seed-averaged m_ψ values at the largest budget z=0.10 in the cycling phase.

z	$m_{\psi\,\mathrm{cyc}}^{\mathrm{rand}}$	$m_{\psi\,\mathrm{cyc}}^{\mathrm{hub}}$	$m_{\psi\,\mathrm{ord}}^{\mathrm{rand}}$	$m_{\psi\,\mathrm{ord}}^{\mathrm{hub}}$
0.00	0.00	0.00	1.00	1.00
0.03	0.02	0.21	0.97	0.95
0.05	0.04	0.40	0.92	0.91
0.08	0.06	0.59	0.88	0.97
0.10	0.08	0.72	0.86	1.00

Cycling phase at z=0.10: hub m_ψ= 0.72 vs random = 0.08 — a **8.9× amplification**. Hubs control *whether* the network orders, not *what* it orders on.

Conclusion. Position beats numbers: the same zealot budget is worth ∼9× more on hubs. But structural leverage only sets the *onset* of order — the cycle still picks the winner (Paper), except at $z \gtrsim 0.09$ in the ordering phase, where hubs saturate their neighbourhoods and finally pin their own strategy (free-node conversion 1.00 at z=0.10). Minority takeover is possible, but only with both structural targeting *and* an already-ordering system.



6.3 Competing Rock+Paper factions (ER, 12 seeds)

What it is. Two opposing zealot factions of equal size z — one locked to Rock, one to Paper — are grown together, and we watch the composition of the free population ($\rho_{\mathrm{rock}}, \rho_{\mathrm{paper}}, \rho_{\mathrm{sciss}}$). In a cyclic game the naive guess is that the two cancel and the third strategy (Scissors, which beats Paper) profits; this experiment checks whether that actually happens.

Parameters — each defined.

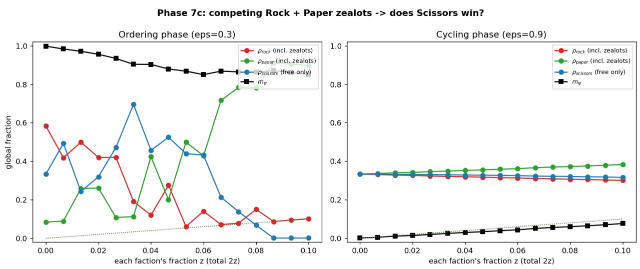
- graphs — ER, N=800, $\langle k \rangle$=10; 12 realisations (seeds 1–12), curves seed-averaged (P5).
- factions — two zealot groups locked to Rock and to Paper respectively, each of fraction z (so $2z$ of the network is committed in total), both placed at random; equal sizes make the cyclic relation the only asymmetry.
- z: 16 points on $[0, 0.10]$ — per-faction fraction, the control variable.
- engine — T=0.65, 1500 sweeps, burn-in 450.
- $\varepsilon \in \{0.3, 0.9\}$ — ordering-phase and cycling-phase panels.
- ρ_x — time-averaged fraction of *all* nodes playing x, zealots included; each faction's own z is therefore a floor on its ρ.

How the numbers are obtained. As Sec. 6.1 but with two locked factions: $\lfloor zN \rceil$ Rock-zealots and $\lfloor zN \rceil$ Paper-zealots drawn from the remaining nodes (both skipped by P1). Each ρ_x is the P2 cluster fraction of strategy x over *all* nodes, time-averaged (P4) and seed-averaged (P5) — so a faction's own z is a floor on its ρ, and the interesting signal is the free remainder.

z (each)	ρ_{rock}	ρ_{paper}	ρ_{sciss} (ord.)
0.00	0.58	0.08	0.33
0.03	0.42	0.11	0.47
0.05	0.06	0.50	0.44
0.08	0.15	0.78	0.07
0.10	0.10	0.90	0.00

With equal Rock+Paper zealots, the ordering-phase population goes to **Paper** (ρ=0.90 at z=0.10), *not* Scissors: Paper is reinforced both by its own zealots and by the Rock-zealots it preys on. Cycling phase stays robust ($m_\psi \leq 0.08$).

Conclusion. Competition between committed factions is decided by the cyclic relation between them, not by their (equal) sizes: the faction that preys on the other collects a double reinforcement and wins outright, while the bystander strategy is eliminated ($\rho_{\mathrm{sciss}} \to 0$). Equal-and-opposite zealotry does not cancel — it amplifies the predator.



6.4 Network defects: edge vs node quenching (ER, 6 seeds)

What it is. Instead of adding agents we *damage* the network: remove a fraction f of edges (broken links) or of nodes (vacancies), then measure how the ε-transition moves. This is the mirror image of Sec. 3.1 — if connectivity stabilises order, quenched disorder should erode it — and comparing the two damage types (matched by the mean degree they leave behind) tests whether only the effective $\langle k \rangle$ matters.

Parameters — each defined.

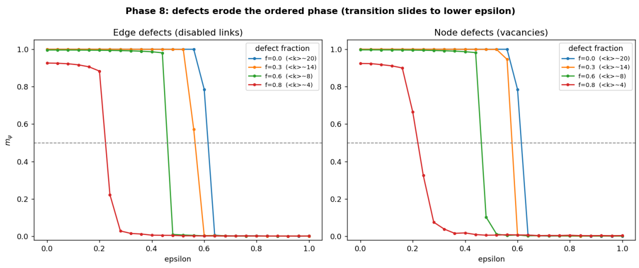
- graphs (before damage) — ER, N=1000 nodes, $\langle k \rangle$=20; a deliberately dense starting point so heavy damage still leaves a connected core; 6 realisations (seeds 1–6), curves seed-averaged (P5).
- damage type $\in$ {edge, node} — the variable under test: remove either links (breaking interactions, keeping all agents) or whole nodes (vacancies, removing agents and all their links).
- $f \in \{0, 0.3, 0.6, 0.8\}$ — fraction removed, uniformly at random; f=0 is the pristine baseline; damage RNG seed = graph seed (quenched: the damage is fixed before the dynamics starts and never heals).
- ε: 26 points on $[0, 1]$ — full transition sweep per (type, f) combination.
- engine — T=0.65, 1000 sweeps, burn-in 300, dynamics seed = graph seed.

How the numbers are obtained. For each (damage type, f): every one of the 6 graphs is damaged, swept in ε (one P1–P4 run per point), and the 6 $m_\psi(\varepsilon)$ curves are averaged (P5); the table's ε_c is P6 applied to that averaged curve. $\langle k \rangle$ after damage $= 2E'/N'$ of the damaged graph (surviving edges E', surviving nodes N'), averaged over the 6 realisations.

f	ε_c(edge)	$\langle k \rangle$(edge)	ε_c(node)	$\langle k \rangle$(node)
0.00	0.61	20.0	0.61	20.0
0.30	0.57	14.0	0.58	14.1
0.60	0.46	8.0	0.46	8.0
0.80	0.22	4.1	0.22	4.0

Removing edges or nodes slides ε_c down ($0.61 \to 0.22$). Edge and node defects **coincide** once matched by the resulting $\langle k \rangle$ $\Rightarrow$ order-stability depends on effective $\langle k \rangle$ only.

Conclusion. *How much* you damage matters; *how* you damage does not. The transition of a damaged network is predictable from a single number — its surviving mean degree — so consensus robustness can be audited by counting links, without knowing which links or nodes were lost.



6.5 Collapse test: damaged $\equiv$ pristine at matched $\langle k \rangle$

What it is. The decisive check on Sec. 6.4: each damaged network contributes one point (resulting $\langle k \rangle$, ε_c), plotted over the *pristine*-ER boundary extracted in Sec. 4.1. If damage only matters through the mean degree it leaves behind, every point must land on the pristine curve.

Parameters — each defined.

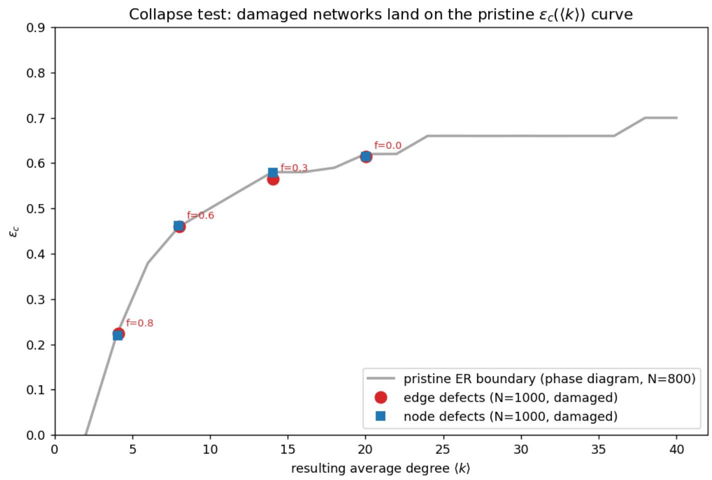
- inputs — no new simulations: reads `defects.csv` (Sec. 6.4, the N=1000 damaged graphs) and `phase_diagram_ER.csv` (Sec. 4, the N=800 pristine graphs); the deliberate N mismatch makes the collapse a stronger test.
- ε_c — linear interpolation of the m_ψ=0.5 crossing (P6), applied identically to both data sets.
- coordinates — each damaged network is placed at its *effective* $\langle k \rangle$, the mean degree $2E'/N'$ left after damage, not at its nominal pre-damage degree.

How the numbers are obtained. Each damaged network contributes the coordinate pair (effective $\langle k \rangle$ after damage, ε_c from P6 on its seed-averaged sweep) — one point per (damage type, f) cell of Sec. 6.4 — and is overplotted on the pristine ER boundary of Sec. 4.1.

quantity	value
$\max_f \lvert\varepsilon_c^{edge} - \varepsilon_c^{node}\rvert$	**0.014**
points on pristine curve	8/8

All 8 damaged-network points (4 edge + 4 node, and at a different N than the pristine sweep) fall on the undamaged boundary: a damaged network is indistinguishable from a pristine one of the same average degree. Data: `defects_collapse.csv`.

Conclusion. Together with Sec. 4.1 (ER $\equiv$ BA within 0.04), this collapses every network studied — random or scale-free, pristine or damaged, N=800 or 1000 — onto a *one-parameter family*: order stability in this model is a function of $\langle k\rangle$ alone. Neither the shape of $P(k)$ nor the damage history leaves a measurable trace at this resolution.



6.6 Robustness: the zealot strategy label — an exact symmetry as a null test

What it is. Secs. 6.1–6.2 lock zealots to Rock. The payoff matrix is exactly symmetric under the cyclic relabeling R→P→S→R, so the choice of label must change *nothing* statistically. The Sec. 6.1 protocol is run for all three labels; this is the designated "parameter that should not matter", tested rather than assumed.

Hypothesis — stated before running. No effect: conversion(z) and $m_\psi(z)$ statistically identical across labels at every z, and the backfire of Sec. 6.1 (the free network adopts the strategy that *beats* the zealots) must appear for every label — Paper-zealots breed Scissors, etc. Any systematic label dependence would expose an implementation bug (asymmetric proposal move, payoff indexing), not physics.

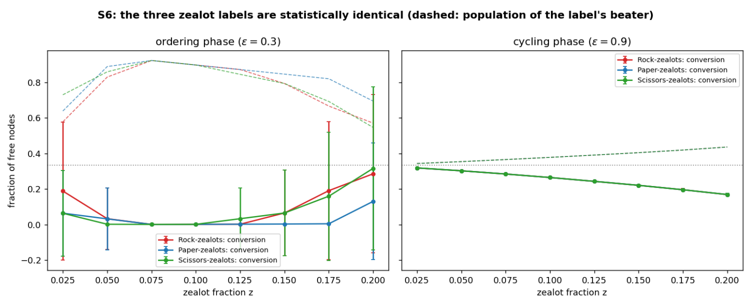
Parameters — each defined.

- zealot strategy $\in$ {R, P, S} — the varied (null) parameter.
- z: 8 fractions on $[0.025, 0.2]$ — z=0 is excluded because with no zealots the label is undefined (all three would measure the *same* run against different names, a deterministic, not statistical, difference).
- seeds — 32 per (label, z, phase); the seed also draws graph and zealot placement, so the placement realisation is varied at the same time.
- system — ER, N=800, $\langle k\rangle$=10, random placement, both phases (ε=0.3 and 0.9): the Sec. 6.1 protocol exactly.
- test — paired permutation test (2000 resamples): under the null the three label values of one seed are exchangeable, so permuting labels *within* each seed gives the exact null distribution of the label spread. (A Gaussian 2σ rule is miscalibrated here: at large z the seed-level outcome is bimodal — a realisation either flips to the zealot consensus or does not.)

How the numbers are obtained. P1–P4 per (label, z, seed, phase) with conversion as in Sec. 6.1; at each z the label spread (max−min of the three label means) is compared to its permutation null. Data: `sensitivity/sens_zealot_symmetry.csv`, p-values in `sens_zealot_pvals.csv`.

	ordering	cycling
z-points with $p < 0.05$	0/8	1/8
min p	0.05	0.02
exactly zero spread	2/8	0/8

1/16 tests significant overall — the textbook false-positive rate for exact null hypotheses at the 0.05 level.



Verdict vs hypothesis. Confirmed, with a bonus. The three labels are statistically indistinguishable (1/16 significant at 0.05 — consistent with pure false positives), the backfire appears for every label (each free network converges onto that label's beater — dashed curves in the figure), and in the cycling phase the three curves overlap to line width. Bonus: at 2 z-points the label spread is *exactly* zero. The engine's RNG consumption is state-independent, so relabeled runs share an aligned random stream; once each reaches its (label-rotated) consensus they are the same trajectory up to rotation, and every label-invariant observable agrees to machine precision (verified directly: the three runs' (r, p, s) are exact cyclic permutations). A per-run exact symmetry check, far stronger than the statistical one.

6.7 Robustness: the damage realisation — does it matter *which* edges die?

What it is. Secs. 6.4–6.5 average over damage realisations. Here one pristine ER graph (N=500, $\langle k \rangle$=20) is damaged 6 independent ways at each of three edge-removal fractions, and each damaged graph gets its own full sweep and ε_c — realisation by realisation, no averaging.

Hypothesis — stated before running. Only *how many* edges die matters, not *which*: per Sec. 6.5, ε_c across damage seeds at fixed f should scatter no more than the surviving effective degree $2E'/N'$ does, and every realisation should land on the pristine $\varepsilon_c(\langle k \rangle)$ boundary at its own effective degree. Scatter should grow mildly with f.

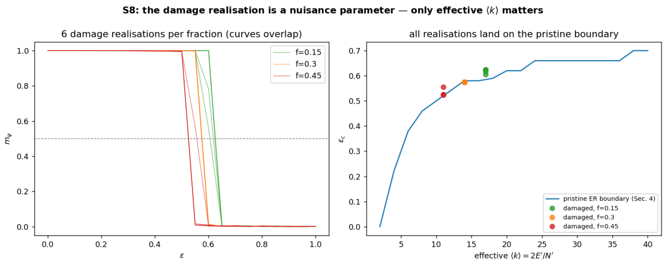
Parameters — each defined.

- damage fraction $f \in \{0.15, 0.3, 0.45\}$ — fraction of edges quenched (removed), as in Sec. 6.4.
- damage seeds $\{1..6\}$ — independent choices of *which* edges die; the varied (null) parameter.
- base graph — one fixed ER realisation (N=500, $\langle k \rangle$=20, graph seed 1), so damage is the only source of variation.
- engine — production protocol (T=0.65, 1500 sweeps, burn-in 30%), ε: 21 points on $[0, 1]$; engine seed = damage seed.
- reference — the pristine ER boundary of Sec. 4.1, interpolated at each realisation's effective $\langle k \rangle = 2E'/N'$.

How the numbers are obtained. Per (f, damage seed): quench, measure $2E'/N'$, run the sweep (P1–P4), extract ε_c (P6); compare seed spread at fixed f and deviation from the interpolated pristine boundary. Data: `sensitivity/sens_defect_seed.csv`.

f	eff. $\langle k \rangle$	ε_c (6 seeds)	dev. from pristine
0.15	17.00±0.00	0.621±0.007	+0.036
0.3	14.00±0.00	0.575±0.000	-0.005
0.45	11.00±0.00	0.530±0.011	+0.010

Effective degree does not even fluctuate: the damage routine removes a deterministic *count* of edges and the giant component survives whole at these densities.



Verdict vs hypothesis. Confirmed. ε_c scatter across damage seeds is at most 0.011 (a quarter of a grid step), every seed-mean sits within 0.036 of the pristine boundary at its effective degree (inside the boundary's own grid resolution), and the scatter grows mildly with f as hypothesised. Sec. 6.5's collapse holds *realisation by realisation*: the residual influence of which edges die is negligible, so damaged networks really are parameterised by the single number $2E'/N'$.

6.8 What the perturbation experiments say together

- **The two phases are vulnerable to different attacks.** The ordered (consensus) phase is *compositionally* fragile — 5% zealots decide which strategy wins (6.1), and two competing factions hand it to the predator (6.3) — but *structurally* predictable: damage moves ε_c only through $\langle k \rangle$ (6.4–6.5). The cycling phase is the mirror image: almost immune to composition attacks ($m_\psi \leq 0.18$ at z=0.20) yet orderable by structural targeting of hubs (6.2).
- ***Whether* and *what* have separate controls.** Whether the population orders is governed by connectivity and placement ($\langle k \rangle$, hubs); *what* it orders on is governed by the cyclic predator logic, which no amount of random zealotry overrides — only hub saturation in an already-ordering system does (6.2).
- **Read as opinion dynamics:** planting stubborn agents backfires (elects the rival that beats you); buying the influencers buys consensus but not *your* consensus; and cutting links erodes consensus by a predictable amount set by the surviving mean degree.

7. Findings summary

Every number below is recomputed from the CSVs at build time (pipeline of Sec. 0.1); section references give the underlying data.

1. **Connectivity stabilises order:** ε_c rises with $\langle k \rangle$ in HMF (0.07→0.69, Sec. 3.1) and in the MC phase diagram (0.00→0.70, Sec. 4).
2. **DMF > HMF:** lower RMSE vs MC on both graphs, advantage ~2.2× larger on heterogeneous BA. Both mean fields overestimate the ordered phase (MC $\varepsilon_c \approx 0.50$ vs MF ≈ 0.62, Sec. 3.2).
3. **Average degree, not $P(k)$:** ER and BA phase boundaries coincide within $\max_k |\Delta\varepsilon_c| = 0.040$ over all 20 degrees (Sec. 4.1).

4. **Genuine transition:** FSS sharpens and ε_c converges to $\approx$0.50; phase portrait shows corner consensus vs limit cycle (Sec. 5).
5. **Zealots back-fire:** a Rock minority flips the free network to Paper (its predator), not Rock (free-Rock conversion 0.00 already at z=0.05, Sec. 6.1) — naive minority takeover fails in a cyclic system.
6. **Hub leverage ∼9×:** hub-placed zealots amplify induced order vs random (cycling phase, z=0.10, Sec. 6.2), but set *whether*, not *what*, the network orders on.
7. **Predator wins under competition:** equal Rock+Paper zealots drive the population to Paper ($\rho = 0.90$, Sec. 6.3), not Scissors.
8. **Defects ≡ effective $\langle k \rangle$:** edge and node damage are equivalent once matched by resulting mean degree (gap ≤ 0.014); ε_c slides 0.61→0.22, and all damaged networks collapse onto the pristine boundary (Secs. 6.4–6.5).
9. **Robustness, tested not assumed:** seeds, ε-grid step, zealot label and damage realisation are verified nuisance parameters (ε_c seed std 0.0025 $\ll$ grid step, Sec. 4.3; label symmetry exact to machine precision where trajectories coalesce, Sec. 6.6; damage realisation scatter $\leq$ 0.011, Sec. 6.7). The regime choices hold with margin: 1500 sweeps is 2× past convergence (Sec. 5.2) and T=0.65 sits in the k-dominated regime (Sec. 4.2).
10. **The transition is first-order-like:** the mean field has a bistable window (consensus + limit cycle coexisting, $[0.631, 0.706]$ at k=10, Sec. 3.4) and the MC pseudo-transition shifts as $1/N$ toward $\varepsilon_c(\infty) \approx 0.58$ at $\langle k \rangle$=20 (Sec. 5.1) — two independent signatures of subcriticality. Quoted ε_c values are finite-N estimates; all cross-network comparisons are at matched N and unaffected.

Generated by `build_report.py` from the regenerated CSVs and `logs/manifest.csv` (31 figures + 37 data tables in the repo). Full narrative in `FINDINGS.md` (root and per folder, incl. `sensitivity/`); run details in `RUN_REPORT.md`; presentation guide in `PRESENTING.md`.